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Application form mySNF

Instrument **Scientific Exchanges**

Part 1: General Information

Basic data

Project Title	Biodiversity Dynamics in Coevolutionary Metaecosystems
Project title in English	Biodiversity Dynamics in Coevolutionary Metaecosystems

Research Field	Life sciences
Main Discipline	30207 Ecology
University	Eidg. Anstalt für Wasserversorgung - EAWAG

Applicant(s)

Main Applicant	**Carlos Melian**

Grant Application

Amount requested (CHF)	Total	**13'500**

Requested starting date	**01.10.2018**
Duration (months)	**1**

Attachments

CV and publication list	CVs_All_Invited_Participants.pdf CV_CarlosMelian.pdf
Further documents for scientific events or research visits	Form_confirmation_self_declaration_Signed.pdf EventProgramme.pdf
Other annexes	ProposalfortheWorkshop.pdf

1. Responsible applicant

Last name	**Melian**
First name	**Carlos**
Function (title)	Group Leader
Academic degree	Dr./PhD
Date of birth	23.08.1969
Gender	male
Civil Status	Married
Language	German
Nationality	Spain
Correspondence address of application	Address of workplace

Home address

Address supplement	
Street, No.	Seestrasse 79
P.O. Box	
Postcode / Zipcode	6047
Place	Kastanienbaum
Country	Switzerland

Address of institute

Name of Institution 1 (e.g. laboratory) *	Eawag
Continuation 1 (e.g. inst /dept.)	
Continuation 2 (e.g. University)	
Street, No.	Seestrasse 79
Address supplement 1 (e.g. building)	
Address supplement 2(e.g. office)	
P.O. Box	
Postcode / Zipcode	6047
Place	Kastanienbaum
State, canton, etc.	
Country	Switzerland

Communication

Secretariat line	
Switchboard	
Direct line	+41 58 765 22 08
Fax office	
Home telephone number	
Cellphone	+41 78 722 88 73
Website	http://www.eawag.ch/about/personen/homepages/melianca/index
E-mail address	carlos.melian@eawag.ch

2. Applicants' employment

Information on employment and function at the anticipated starting date of the grant

Name	**Melian, Carlos**
Employment at the anticipated starting date of the grant	permanent contract
Level of employment %	100
Function in the context of this grant application	Head of (e.g. institute, department, center, clinic)
Professorship	Assistant professor with tenure track
Doctorate (PhD)?	Yes
Date of doctorate (PhD)	01.02.2005
PhD supervisor	
Country of doctorate	Spain
Remarks	
Further employments	

3. Basic data I

Original title	Biodiversity Dynamics in Coevolutionary Metaecosystems
Title in English	Biodiversity Dynamics in Coevolutionary Metaecosystems
Requested starting date	01.10.2018
Duration (months)	1
Research field	Life sciences
Main discipline	30207 Ecology
Sub-discipline(s)	30206 Environmental Research

4. Basic data II

Summary

Background and Rationale
Joining theoretical and empirical evolutionary ecologists to develop and test theoretical models that incorporate the empirical patterns of reciprocal trait changes will be critical to improve our predictions on biodiversity dynamics and their responses to multiple disturbances in species- and interaction-rich ecological communities. The aim of the workshop is to integrate trait-species-space interactions data into new process-based coevolutionary metaecosystem models. Such integration might facilitate the exploration of patterns of coexistence of co-evolved species, their role in ecosystem functioning, and the risk of collapse of co-evolved ecosystems across different spatio-temporal scales, which is of critical importance for three reasons. First, the extinction of species caused by human-generated disturbances occurs at an unprecedented rate, and yet most models predicting extinction rates often assume that species evolve independently of one another. Second, although eco-evolutionary approaches of species interactions are increasingly common, studies connecting co-evolving trait distributions within and between species at small scales to investigate the large-scale association between coevolutionary hot spots, biodiversity function and meta-ecosystem dynamics are rare. Third, although many experiments and empirical studies have shown convergence of time scales between ecological and coevolutionary processes, there is a gap in understanding how ecological and coevolutionary dynamics in species-rich assemblages shape biodiversity and ecosystem function in a spatio-temporal context.

Objectives and Methods
We will focus on implementing analytic tools and testing them with empirical patterns from data sets containing many species, sites and traits. We propose to bring together biologists skilled in theoretical ecology, evolutionary biology, and community ecology to discuss about the following two questions:
Question 1: How does reciprocal trait change influence biodiversity dynamics, species extinctions and ecosystem function in species-rich meta-ecosystems?
We aim to integrate the effects of ecological, coevolutionary, and migration dynamics in shaping biodiversity in meta-ecosystems. We will merge the meta-ecosystem perspective into existing models of coevolutionary interaction networks. We will implement scenarios to predict biodiversity distribution across space and time under different functional interactions (i.e., trait matching and exploitation barriers, assuming weak and strong reciprocal selection strengths. Metacommunity and meta-ecosystem theory approaches will complement the functional scenarios to understand how spatial processes governed by dispersal or niche width can alter the coevolutionary dynamics within communities by contributing to species trait variation and network structure and stability, and thus ecosystem functioning.
Question 2: How do the spatial arrangement of ecosystems and coevolutionary dynamics interact to facilitate rapid evolutionary changes in meta-ecosystems?
Most interactions between species generate reciprocal selection pressures, and the coevolutionary dynamics that emerges over time varies with the fitness consequences of the ecological interaction (i.e., competitive, mutualistic, or antagonistic) and the architecture of the coevolving interactions between species. Yet, how the relationship between species traits and coevolution at local scales and the arrangement of ecosystems at the landscape scale produce rapid biodiversity changes and coevolutionary hot spots is not well understood. We will run simulations under landscapes with different spatial arrangements and migration scenarios to disentangle the interplay between coevolutionary dynamics at the local and the landscape level.

	Expected results and Impact We see great potential in linking the research of the participants to produce new synthesis in merging coevolution, interaction networks and meta-ecosystems. Specifically, the workshop aims to decipher the interplay between coevolution in local ecosystems for antagonistic and mutualistic interactions with the effects of migration under different landscape arrangements. This is a venue that can move forward the needed synthesis between the ecological and evolutionary processes that shape biodiversity. Furthermore, our understanding of coevolution among many interacting species at different spatial scales and its effects on biodiversity dynamics and ecosystem function is just starting to emerge. Yet, the integration between robust empirical patterns and process-based models, the main aim of this workshop, remain at a very incipient stage.
Keywords	Biodiversity dynamics
	Metacommunity theory
	Coevolution
	Theoretical ecology
	Biodiversity patterns
Language of correspondence	German
Financial administration	EAWAG Verwaltung zHv Frau Barbara Breu Rütti

5. University or research institution

University	Eidg. Anstalt für Wasserversorgung - EAWAG
Remarks	

6. Requested funding

Requested funding	**Total (CHF)**	**Year 1**
Total (CHF)	**13'500**	**13'500**

Research funds	**Total (CHF)**	**Year 1**
Travel & subsistence allowances	13'500	13'500
Total (CHF)	**13'500**	**13'500**
Total (%)	**100%**	**100%**

Details

Travel & subsistence allowances	**Total (CHF)**	**Year 1**
Andreazzi, Cecilia, Associate Researcher, Fundação Oswaldo Cruz, Rio de Janeiro, Brasil	2'360	2'360
Relation to research plan / Comments / Additions: Airfare (Brasil) 1750CHF Food (6 days) 350CHF Train Airport-KB round trip 90CHF Daily bus pass 50CHF 6 nights 110CHF Linen 10CHF		
Arim, Matias, Full Professor, University of the Republic, Maldonado, Uruguay	2'380	2'380
Relation to research plan / Comments / Additions: Airfare (Uruguay) 1770CHF Food (6 days) 350CHF Train Airport-KB round trip 90CHF Daily bus pass 50CHF 6 nights 110CHF Linen 10CHF		
Astegiano, Julia, Researcher, CONICET, Cordoba, Argentina	2'360	2'360
Relation to research plan / Comments / Additions: Airfare (Argentina) 1750CHF Food (6 days) 350CHF Train Airport-KB round trip 90CHF		

	Daily bus pass 50CHF 6 nights 110CHF Linen 10CHF		
Becks, Lutz, Researcher, Max-Planck Inst., Plön, Germany		1'010	1'010
Relation to research plan / Comments / Additions	Airfare (Germany) 400CHF Food (6 days) 350CHF Train Airport-KB round trip 90CHF Daily bus pass 50CHF 6 nights 110CHF Linen 10CHF		
Heleno, Ruben, Auxiliary Researcher, University of Coimbra, Portugal		1'010	1'010
Relation to research plan / Comments / Additions	Airfare (Portugal) 400CHF Food (6 days) 350CHF Train Airport-KB round trip 90CHF Daily bus pass 50CHF 6 nights 110CHF Linen 10CHF		
Huang, Weini, Lecturer, Queen Mary Univ. London, London, UK		1'010	1'010
Relation to research plan / Comments / Additions	Airfare (UK) 400CHF Food (6 days) 350CHF Train Airport-KB round trip 90CHF Daily bus pass 50CHF 6 nights 110CHF Linen 10CHF		
Massol, François, Researcher, CNRS, Lille, France		1'010	1'010
Relation to research plan / Comments / Additions	Airfare (France) 400CHF Food (6 days) 350CHF Train Airport-KB round trip 90CHF Daily bus pass 50CHF 6 nights 110CHF Linen 10CHF		
Sabatino, Malena, Researcher CONICET, Mar del Plata, Argentina		2'360	2'360
Relation to research plan / Comments / Additions	Airfare (Argentina) 1750CHF Food (6 days) 350CHF Train Airport-KB round trip 90CHF Daily bus pass 50CHF 6 nights 110CHF Linen 10CHF		
Total (CHF)		**13'500**	**13'500**
Total (%)		**100%**	**100%**

7. Timeframe of the scientific exchange

Starting date	15.10.2018
Place	Kastanienbaum, Switzerland
Duration (days)	5

8. Contents

General information

Aim and motivation	Motivation and AimCoevolution, i.e. changes of species traits driven by reciprocal selection within and among interacting species, has been invoked as a primary mechanism generating much of Earth's biodiversity. Two main theories have been proposed to explain the dynamics of coevolutionary diversification. The Fisherian sexual selection theory, which states that female and male traits can lead to sexual isolation and speciation, and the Geographic mosaic theory of coevolution, which suggests that reciprocal changes in traits determining species interaction produce coevolutionary diversification as interspecific interactions control reproductive isolation and speciation. Despite the potential of such theories for explaining much of Earth's biodiversity, understanding how coevolution of intra- and inter-specific traits accounting for ecological interactions alter ecosystem dynamics across heterogeneous

	landscapes remain elusive. Such understanding is particularly important in ecosystems with species- and interaction-rich ecological networks because global changes might trigger many species extinctions and the collapse of ecosystems. The aim of the workshop is to integrate trait-species-space interactions data into new process-based coevolutionary meta-ecosystem models to understand the impact of coevolution on the dynamics of biodiversity. We will focus on implementing analytic tools and testing them with empirical patterns from data sets containing many species, sites and traits. We propose to bring together biologists skilled in theoretical ecology, evolutionary biology, and community ecology to discuss about the following two questions:1: How does reciprocal trait change influence biodiversity dynamics, species extinctions and ecosystem function in species-rich meta-ecosystems? 2: How do the spatial arrangement of ecosystems and coevolutionary dynamics interact to facilitate rapid evolutionary changes in meta-ecosystems?Added value for Swiss-researchThe workshop will be a one-off event, and it will lead to new international collaborations between Swiss institutions and institutions from abroad. One of the key milestones of the workshop will be to draft new ideas to produce a grant proposal draft to be submitted to the H2020, Biodiversa, and/or sDiv working groups. We also see great potential in linking the research of the participants based in Switzerland to produce new synthesis in merging coevolution, interaction networks and meta-ecosystems. Specifically, deciphering the interplay between coevolution in local ecosystems for antagonistic and mutualistic interactions with the effects of migration under different landscape arrangements is a venue that can move forward the needed synthesis between the ecological and evolutionary processes that shape and maintain biodiversity. Furthermore, our understanding of coevolution among many interacting species at different spatial scales and its effects on biodiversity dynamics and ecosystem function is just starting to emerge. Yet, the integration between robust empirical patterns and process-based models, the main aim of this workshop, is currently missing.Expected results and communication of results We will work on an opinion, review or ideas-perspectives paper entitled “Biodiversity dynamics and ecosystem functioning in coevolutionary meta-ecosystems” for submission to Trends in Ecology and Evolution, Ecology Letters or Biological reviews. Prior to the meeting, attendees will provide written perspectives about the main overarching questions in their fields and the trait-species-spatial interaction data they can provide. On the first day, participants will present these ideas based on their own research programs. On subsequent days we will synthesize these perspectives into a single review/opinion and synthesis style paper integrating the datasets to the methods that we will develop and implement during the workshop and following the planned research visit.
Organisation	The workshop will be hosted at Eawag’s Center for Ecology, Evolution & Biogeochemistry (CEEB), in Kastanienbaum, Switzerland. This is an ideal venue because all the invited guests can stay on-site in Eawag apartments, directly on the shore of Lake Lucerne. The meeting facilities are very suitable and have been previously used to host several small workshops and student courses with great success. Such events also help to raise our international profile, and, in the past, have led to repeat visits from graduate students and visiting faculty on sabbatical terms. Eawag will provide the necessary infrastructure for the workshop (venue, consumables, IT, etc.). No other funds are needed to support this workshop.

Target audience

Expected number of people (total number of participants)	14
Remark	We have invited 8 scientists from abroad (7 countries) and 6 from Switzerland working in different departments and institutions (Eawag, Ecology and Fish Ecology and Evolution department) and WSL.

Active participants

Number of active participants from the academic field	7
Number of active participants from outside	7

the academic field	
Remark	We have selected a balanced group of participants representing a wide range of scientific ages, expertise, and levels of experience from the academic field object of the proposal (7) and outside the academic field (7). We have identified a mixture of people with (i) skills in mathematical biology, (ii) experience with software development, and (iii) strong conceptual skills in evolutionary biology, meta-ecosystem theory, model systems and biodiversity patterns from field studies.
Number of women	7
Number of PhD students	1
Number of students	0
Remark	We have carefully chosen on behalf of gender equality (7F:7M, from 7 countries), to represent diverse fields (e.g. theoretical ecologists, evolutionary biologists, and community ecologists) and scientific career stages (2 Profs., 5 Associate Prof. or group leaders (2 tenure track), 4 research scientists, 2 postdoctoral researches, and 1 PhD student).

9. General remarks on the project

Subject	Submitted in parallel with a research visit
Communication	This scientific event will be joined to a research visit submitted in parallel.
Confidential	No

Cecilia Siliansky de Andreazzi
04/24/1984. Rio de Janeiro, RJ, Brazil

Curriculum Vitae

Academic address Fundação Oswaldo Cruz, Presidência da Fiocruz, Campus Fiocruz Mata Atlântica.
Estrada Rodrigues Caldas 3400 Pavilhão Zanine
Taquara
22713375 - Rio de Janeiro, RJ - Brasil

E-mail: cecilia.andreazzi@fiocruz.br, ci.andreazzi@gmail.com

Website: http://lattes.cnpq.br/7344032911778389

Profile

Associate researcher at Fundação Oswaldo Cruz with interest in the ecology and evolution of species interaction networks. Background in ecology, disease ecology, evolutionary biology, complex systems, mathematical modeling and numerical simulations.

Education

2012 - 2016 Ph.D in Ecology.
Universidade de São Paulo, USP, São Paulo, Brasil
Title: Coevolution in antagonistic species interaction networks: structure and dynamics
Advisor: Paulo Roberto Guimarães Júnior
Scholarship: Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)

2015 - 2016 Visiting Ph.D student (6 months)
Swiss Federal Institute of Aquatic Science and Technology
Advisor: Carlos Melián
Scholarship: Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)

2006 - 2008 M.Sc in Ecology.
Universidade Federal do Rio de Janeiro, UFRJ, Rio de Janeiro, Brasil
Title: Reproductive phenology and seed dispersal and predation of *Attalea humilis* palm in Atlantic Forest fragments
Advisors: Fernando Antonio dos Santos Fernandez and Alexandra Pires
Scholarship: Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)

2002 - 2006 B.Sc in Biological Sciences.
Universidade Federal do Rio de Janeiro, UFRJ, Rio de Janeiro, Brasil
Title: Small mammals population ecology in the Pantanal
Advisor: Paulo Sérgio D'Andrea
Scholarship: Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)

Appointments

2006 - 2016 Research assistent, Fundação Oswaldo Cruz (FIOCRUZ, Rio de Janeiro, Brazil)
2016 - Associate researcher, Fundação Oswaldo Cruz (FIOCRUZ, Rio de Janeiro, Brazil)

Teaching

2017 - *Novel methods for the study of Biodiversity,* at the Instituto Oswaldo Cruz graduate programs (annually offered), Brazil
2017 - *Population Ecology*, at the Instituto Oswaldo Cruz graduate programs (annually offered), Brazil

Juries

Examiner at 1 PhD jury, 2 PhD qualifications and 2 Bachelor juries.

Complementary Education

2014 - 2014 Complex Systems Summer School
Santa Fé Institute, New Mexico, United States
2013 - 2013 II Southern-Summer School on Mathematical Biology.
ICTP South American Institute for Fundamental Research, ICTP-SAIFR, Brazil
2013 - 2013 Energetic Approach to Food Webs.
ICTP South American Institute for Fundamental Research, ICTP-SAIFR, Brazil
2012 - 2012 Food Webs Dynamics.
Laboratório Nacional de Computação Científica, LNCC, Petrópolis, Brazil
2007 - 2007 III Ibero-American Course on Frugivory and Seed Dispersal
Universidade Estadual Paulista, UNESP RIO CLARO, Brazil

Peer Review

- International Journal of Biodiversity
- Cadernos de Saúde Pública
- Chiroptera Neotropical
- Proceedings of the Royal Society – B
- Oecologia
- Oikos
- Ecography
-

Grants & Awards

2016 *Best poster***,** Conference on Mathematical Modeling and Control of Communicable Diseases
2012 - CNPq - Ph.D. Scholarship
2006 - 2007 CNPq - M.Sc. Scholarship
2006 Research Grant, IdeaWild
2005 *The ten best Scientific Initiation works in FIOCRUZ*, Fundação Oswaldo Cruz.
2003 - 2006 CNPq - Undergraduate Research Scholarship

Publications

1. MELIÁN, C. J., MATTHEWS, B., **ANDREAZZI, C. S.,** RODRIGUEZ, J. P., HARMON, L. J., FORTUNA, M. A. 2018. Deciphering the Interdependence Between Ecological and Evolutionary Networks. *Trends in Ecology & Evolution*. In press.
2. **ANDREAZZI, C. S.,** GUIMARÃES Jr., P.R., MELIÁN, C. J. 2018. Eco-evolutionary feedbacks promote fluctuating selection and long-term stability of antagonistic networks. *Proceedings of the Royal Society B-Biological Sciences*: 285: 20172596.
3. STELLA, M., SELAKOVIC, S., ANTONIONI, A., **ANDREAZZI, C. S.** 2018**.** Ecological multiplex interactions determine the role of species for parasite spread amplification. *eLife* 7:e3.

4. RAIMUNDO, R. L. G., MARQUITTI, F. M. D., **ANDREAZZI, C. S.,** PIRES, M. M., GUIMARÃES Jr., P. R. 2017. Ecology and evolution of species-rich interaction networks**.** In Dáttilo, W. and Rico-Gray, V. (Eds.). Book: Ecological Networks in the Tropics. Springer Publisher.
5. **ANDREAZZI, C. S**., THOMPSON, J., GUIMARÃES Jr. P. R. 2017. Network structure and selection asymmetry drive coevolution in species-rich antagonistic interactions. *The American Naturalist* 190: 99-115.
6. STELLA, M., **ANDREAZZI, C. S**., SELAKOVIK, S., ALIREZA, G., ANTONIONI, A. 2017. Parasite Spreading in Spatial Ecological Multiplex Networks. Journal of Complex Networks 5(3): 485-511.
7. **ANDREAZZI, C. S**., PIRES, A. S., PIMENTA, C. S., FERNANDEZ, F. A. S. 2012. Increased female reproduction favours the large-seeded palm *Attalea humilis* in small Atlantic Forest fragments. *Journal of Tropical Ecology* 28: 321-325.
8. **ANDREAZZI, C. S**., PIMENTA, C. S., PIRES, A. S., FERNANDEZ, F. A. S., OLIVEIRA-SANTOS, L. G. & MENEZES, J. F. S. 2012. Increased Productivity and Reduced Seed Predation Favor a Large-seeded Palm in Small Atlantic Forest Fragments. *Biotropica*, 44: 237-245.
9. **ANDREAZZI, C. S**., RADEMAKER, V., GENTILE, R., HERRERA, H., JANSEN, A. M., DANDREA, P. S. 2011. Population ecology of small rodents and marsupials in a semi-deciduous tropical forest on the Southeast Pantanal wetland, Brazil. *Zoologia* 28: 762-770.
10. MORATELLI, R., **ANDREAZZI, C. S**., OLIVEIRA, J. A. & CORDEIRO, J. L. P. 2011. Current and potential distribution of *Myotis simus* (Chiroptera, Vespertilionidae). *Mammalia*, 75: 227-234.
11. **ANDREAZZI, C. S**., PIRES A. S. & FERNANDEZ, F. A. S. 2009. Mamíferos e palmeiras neotropicais: interações em paisagens fragmentadas. *Oecologia Brasiliensis* 13: 554-574.

Scientific Communication

1. **ANDREAZZI, C. S**., ASTEGIANO, J., GUIMARÃES Jr. P. R. Mechanisms driving the evolution of species-rich interaction networks In: **International Conference of the Institute for Biocomputation & Physics of Complex Systems**, BIFI**,** Zaragoza. 2018.

2. **ANDREAZZI, C. S**., ASTEGIANO, J., GUIMARÃES Jr. P. R. Coevolution by trait matching and exploitation barriers shape the structure of antagonistic and mutualistic networks in different ways In: **Ecological Society of America meeting,** Portland. 2017.

3. **ANDREAZZI, C. S**., ASTEGIANO, J., GUIMARÃES Jr. P. R. Evolução e Co-evolução em redes antagonistas E mutualistas In: **VI Reunión Binacional de Ecologia,** Puerto Iguazú. 2016.

4. **ANDREAZZI, C. S**., GUIMARÃES Jr. P. R., MELIÁN, C. J. Eco-evolutionary feedbacks promote interaction fluctuating selection and long-term stability of antagonistic networks. In: **Evolution meeting,** Austin. 2016.

5. STELLA, M., **ANDREAZZI, C. S**., SELAKOVIK, S., ALIREZA, G., ANTONIONI, A. Parasite Spreading in Spatial Ecological Multiplex Networks**.** In: **Conference on Mathematical Modeling and Control of Communicable Diseases**, Rio de Janeiro. 2016.

6. **ANDREAZZI, C. S**., GUIMARÃES, P. R., Jr. Generalist enemies reduce the escalation of attack and defense traits in species-rich networks**.** In: **Ecological networks: theory, empiricism and practice**

in a changing world, Bristol. 2015.

7. **ANDREAZZI, C. S**., GUIMARÃES, P. R., Jr. Network structure drive coevolutionary dynamics in species-rich antagonistic interactions**.** In: **Evolution meeting**, Guarujá. 2015.
8. **ANDREAZZI, C. S**., GUIMARÃES, P. R., Jr. Coevolution in antagonistic networks**.** In: **Winter Workshop on Complex Systems**, Brussels. 2015.
9. **ANDREAZZI, C. S.**, CORDEIRO, J. L. P., ABREU, A. V. G., FRAGOSO, J. M. V., Oliveira, L. F. B. Experimental evaluation of *Attalea phalerata* seed predation along a gradient of palm density in the Brazilian Pantanal In: **49th Annual Meeting of the Association for Tropical Biology and Conservation**, Bonito, 2012, Bonito.
10. **ANDREAZZI, C. S.**, CORDEIRO, J. L. P., MORATELLI, R. Distribution models for three phylogenetically close related South American species of *Myotis* Kaup (Chiroptera, Vespertilionidae) In: **10th International Mammalogical Congress**, Mendoza, 2009,
11. **ANDREAZZI, C. S**., PIMENTA, C. S., PIRES, A. S., FERNANDEZ, F. A. S. Dispersão de frutos da palmeira pindoba (Attalea humilis) por mamíferos In**: IV Congresso Brasileiro de Mastozoologia**, São Lourenço, 2008.
12. PIRES, A. S., **ANDREAZZI, C. S.,** PIMENTA, C. S., FERNANDEZ, F. A. S., GALETTI, M. Seed dispersal and predation in two Atlantic Forest Palms with different responses to habitat loss In: **Annual Meeting of the Association for Tropical Biology and Conservation: Linking Tropical Biology with Human Dimensions**, Morelia, 2007.
13. **ANDREAZZI, C. S.,** PIMENTA, C. S., PIRES, A. S., FERNANDEZ, F. A. S. Predação de sementes de Attalea humilis Mart. Ex. Spreng (Arecaceae) em fragmentos de Mata Atlântica (RJ) de diferentes tamanhos In**: VIII Congresso de Ecologia do Brasil**, Caxambú, 2007.
14. **ANDREAZZI, C. S.,** RADEMAKER, V., JANSEN, A. M., DANDREA, P. S. Ecologia de Populações de pequenos mamíferos numa área preservada do pantanal da Nhecolândia In: **III Congresso Brasileiro de Mastozoologia**, Aracruz, 2005.

CURRICULUM VITÆ: FRANÇOIS MASSOL

Unit Evo-Eco-Paléo, SPICI group
CNRS UMR 8198, Université Lille, Bâtiment SN2
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Born September 17th, 1979
French nationality
Single, three children
francois.massol@m4x.org

Personal webpage: https://sites.google.com/a/polytechnique.org/francoismassol/home
Research group webpage: http://spici.weebly.com/
RESEARCHGATE; MENDELEY; ACADEMIA; ORCID; RESEARCHER ID; GOOGLE SCHOLAR ; SCOPUS

Recent professional positions

2012 – **Permanent researcher** CNRS, CEFE Montpellier (2012 – 2014), EEP Lille (2013 –)
Hired through a national competition, with ~6 posts for ~200 junior applicants in ecology/evolution.

2009 – 2010 **Marie Curie post-doc** at the University of Texas, Austin, USA (supervisor: M. Leibold)

2008 – 2012 **Permanent researcher** IRSTEA, Aix-en-Provence (2008 – 2009 and 2010 – 2012)

2002 – 2008 **Engineer / Civil servant** ENGREF, Paris

Education

2015 **Habilitation** (HDR) in Biodiversity Science at the University of Lille
HDR is the diploma allowing researchers to be PhD student main supervisors; ~ tenure or privatdozent.

2004 – 2008 **PhD** in Population Biology & Ecology at the University of Montpellier
Supervised by D. Gerdeaux and P. Jarne; paid as civil servant for the Ministry of Agriculture.

2003 – 2004 **Master** in Evolutionary Biology & Ecology at the University of Montpellier

1999 – 2003 **Interdisciplinary graduate studies** at Ecole Polytechnique and ENGREF, 2 MSc degrees

Advising & supervision

2013 – Member of 8 PhD advisory committees (Montpellier, Toulouse, La Réunion, Marseille, Dijon)

2008 – Adviser or co-adviser of 7 post-docs, 3 PhD students, 14 MSc students
1 PhD student defended his thesis (F. Laroche) and is now a permanent researcher in France.
5 former post-docs now have permanent or tenure-track positions in Europe, Argentina or the US (K. Allhoff, J. Astegiano, J. Pantel, M. Pautasso, M. Thomas).

Teaching

2014 **Invited graduate course** – evolutionary dynamics, SAIFR, São Paulo, Brazil

2013 **Invited graduate course** – metacommunity theory, ETH/Uni. Zürich, Ascona, Switzerland

2013 – **Regular lectures at MSc level** (10 – 16h per year) – ecological modelling, evolutionary biology, methods for the study of evolution, the publication process, etc., Univ. Lille

Funding

As principal investigator (PI)

2009 – 2011	**DEFTER**	Post-doctoral fellowship (EU Marie Curie programme) on "dispersal effects on food web functioning: theory and experimental research based on freshwater plankton systems" (in M. Leibold's laboratory in Austin, TX)
2010 – 2012	**FoodWebs**	Working group at the National Institute for Mathematical and Biological Synthesis (NIMBioS) on "food web dynamics and stoichiometric constraints in metaecosystems" (PIs: C. Klausmeier, M. Leibold, F. Massol & R. Sterner)
2013 – 2018	**MIRES**	Working group funded by the Réseau National Systèmes Complexes (RNSC) and the Institut National de la Recherche Agronomique (INRA) applied mathematics and informatics department on "interdisciplinary methods for the study of seed exchange networks" (PIs: P. Barbillon, S. Caillon, F. Massol, M. Thomas & N. Verzelen)
2013 – 2014	**DyBRES**	Working group funded by the RNSC on "biodiversity dynamics in spatial ecological networks" (PIs: S. Manel, F. Massol & F. Munoz)
2014 – 2017	**COREIDS**	Working group at the Centre de Synthèse et d'Analyse de la Biodiversité (CESAB) on "Predicting community resilience to invasions from diversity and network structure" (PIs: P. David & F. Massol)
2014	**RESEDA**	Bilateral cooperation (funded by the University of Lille 1) with the University of Exeter on "selection regimes, dispersal and local adaptation" (PIs: F. Débarre & F. Massol)

2014 – 2018	**ARSENIC**	Project funded by the Agence Nationale de la Recherche (ANR) on "adaptation and resilience of spatial ecological networks to human-induced changes" (PIs: N. Loeuille & F. Massol)
2016 – 2017	**FEEDING**	Interregional project Ghent-Louvain-Lille on "functional and evolutionary ecology of dispersal: integrative approaches under global change" (PIs: N. Schtickzelle, D. Bonte & F. Massol)
2017 – 2018	**E²-SPACE**	Interregional project Ghent-Louvain-Lille on "eco-evolutionary dynamics in space" (PIs: N. Schtickzelle, D. Bonte & F. Massol)

Others

2011 – 2014	**NETSEED**	Working group at the CESAB on "seed exchange networks" (PI: D. McKey)
2012 – 2017	**AFFAIRS**	Project funded by the ANR BIOADAPT programme on "adaptation of fragmented freshwater assemblages in rapidly-changing environments" (PI: P. David)
2015 – 2018	**AREOLAIRE**	Project funded by the Région Nord Pas-de-Calais / FRB programme on "adaptation, regression and expansion at species range margins" (PI: A. Duputié)
2015 – 2016	**MADRES**	Project funded by the CNRS interdisciplinary mission MoMIS programme on "modelling and analysis of the dynamics of seed exchange networks" (PI: S. Martin)
2018 – 2019	**sEcoEvo**	Working group at the sDiv synthesis centre (Berlin, Germany) on "Biodiversity Dynamics – The Nexus Between Space & Time" (PIs: R. G. Gillespie & M. Hickerson)
2018 – 2021	**SUBANTECO**	Project funded by the Institut polaire français Paul-Emile Victor (IPEV) on "Subantarctic biodiversity, effects of climate change and biological invasions on terrestrial biota" (PI: D. Renault)
2017 – 2021	**NGB**	Project funded by the ANR on "next-generation biomonitoring of change in ecosystem structure and function" (PI: D. Bohan)

Awards

2016 – 2020 Supervision and Research Bonus (PEDR), granted by the CNRS

Commissions of trust

2017 –	**Founder, manager and recommender** for *Peer Community in Ecology*
2017 –	**Participation in researcher and lecturer hiring committees** (Univ. Paris Diderot, IRSTEA)
2016 – 2017	**Guest volume editor** for the 56th and 57th volumes of *Advances in Ecological Research*
2015 –	**Member of the administrative board** of the French Society of Ecology and Evolution (SFE²)
2015 –	**Referee** at 1 HDR defence; **examiner** at 2 HDR defences (Montpellier, Toulouse, Lille)
2014 –	**Referee** at 1 PhD defence; **examiner** at 5 PhD defences (Paris, Dijon, Montpellier, Ecole Polytechnique)
2014 –	**Subject editor** for the journal *Oikos*
2011 –	**Reviewer** of proposals for funding agencies (Austria, Belgium, France, Israel, Russia, Switzerland)
2008 –	**Reviewer** for > 20 journals including *PNAS, Proc. R Soc B, Ecol. Lett., Nat. Commun., Nat. Ecol. Evol.*

Since 2007, I have authored **55 peer-reviewed articles** (**14 as first author, 9 as last author**), in major multidisciplinary (1 *Nature*, 2 *Nature Communications,* 3 *Scientific Reports*, 1 *Science Advances*) and general biology, ecology and evolutionary biology journals (2 *Trends in Ecology and Evolution*, 4 *Ecology Letters*, 1 *Biological Reviews*, 1 *Ecological Monographs*, 1 *Methods in Ecology and Evolution*, 2 *Proceedings of the Royal Society B*, 4 *Evolution*, 1 *Journal of Ecology*...). I have also contributed to **1 book chapter** and edited **2 books** (volumes 56 and 57 of the *Advances in Ecological Research* series). I gave **2 invited plenary lectures**, **7 invited oral communications**, **5 oral contributions**, 21 oral presentations in workshops and 16 seminars in various universities.

Citations: 1682 (Google Scholar) / 1104 (ISI); **1 highly cited paper** (on ISI web of science[1])
H-index: 22 (Google Scholar) / 18 (ISI)

All articles can be found here: https://sites.google.com/a/polytechnique.org/francoismassol/publications
In the following, students are underlined. * indicates co-first authorship. Impact factors (IF) were obtained from the 2016 ISI Journal Citation Reports, numbers of citations were obtained from Web of Science on 25/05/2018.

[1] received enough citations to place them in the top 1% of their academic field based on a highly cited threshold for the field and publication year.

Representative peer-reviewed publications (non-exhaustive list)

(1) Gravel, D.*, **Massol**, **F.***, & Leibold, M. (2016) Stability and complexity in model meta-ecosystems. *Nature Communications,* **7**, 12457. **11 citations**; **journal IF: 12.124**

(2) **Massol**, **F.** & Débarre, F. (2015) Evolution of dispersal in temporally variable environments: the importance of life cycles. *Evolution*, **69**, 1925-1937. **5 citations**; **journal IF: 4.201**

(3) **Massol, F.**, Dubart, M., Calcagno, V., Cazelles, K., Jacquet, C., Kéfi, S., & Gravel, D. (2017) Island biogeography of food webs. *Advances in Ecological Research*, **56**, 183-262. **7 citations**; **journal IF: 5.056**

(4) **Massol**, **F.**, Gravel, D., Mouquet, N., Cadotte, M. W., Fukami, T., & Leibold, M. A. (2011) Linking ecosystem and community dynamics through spatial ecology. *Ecology Letters*, **14**, 313-323. **99 citations**; **journal IF: 9.449**

(5) Gravel, D., **Massol**, **F.**, Canard, E., Mouillot, D., & Mouquet, N. (2011) Trophic theory of island biogeography. *Ecology Letters*, **14**, 1010-1016. **77 citations**; **journal IF: 9.449**

(6) **Massol**, **F.**, Duputié, A., David, P., & Jarne, P. (2011) Asymmetric patch size distribution leads to disruptive selection on dispersal. *Evolution*, **65**, 490-500. **32 citations**; **journal IF: 4.201**

(7) Laroche, F., Jarne, P., Lamy, T., David, P., & **Massol**, **F.** (2015) A neutral theory for interpreting correlations between species and genetic diversity in communities. *American Naturalist*, **185**, 59-69. **14 citations**; **journal IF: 4.167**

(8) Tasiemski, A., **Massol**, **F.**, Cuvillier-Hot, V., Boidin-Wichlacz, C., Roger, E., Rodet, F., Fournier, I., Thomas, F., & Salzet, M. (2015) Reciprocal immune benefits based on the complementary production of antibiotics by the leech *Hirudo verbana* and its gut symbiont *Aeromonas veronii*. *Scientific Reports*, **5**, 17498. **7 citations**; **journal IF: 4.259**

(9) Cheptou, P.-O. & **Massol**, **F.** (2009) Pollination fluctuations drive evolutionary syndromes linking dispersal and mating system. *American Naturalist*, **174**, 46-55. **54 citations**; **journal IF: 4.167**

(10) Duputié, A. & **Massol**, **F**. (2013) An empiricist's guide to theoretical predictions on the evolution of dispersal. *Interface Focus*, **3**. **31 citations**; **journal IF: 2.693**

(11) Calcagno, V.*, **Massol**, **F.***, Mouquet, N., Jarne, P., & David, P. (2011) Constraints on food chain length arising from regional metacommunity dynamics. *Proceedings of the Royal Society Biological Sciences Series B*, **278**, 3042-3049. **25 citations**; **IF: 4.940**

(12) Schleuter, D., Daufresne, M., **Massol**, **F.**, & Argillier, C. (2010) A user's guide to functional diversity indices. *Ecological Monographs*, **80**, 469-484. **236 citations**; **IF: 8.759**

Other representative publications (non-exhaustive lists)

Open review publications

Lavoir, A.-V., Staudt, M., Schnitzler, J. P., Landais, D., **Massol**, **F.**, Rocheteau, A., Rodriguez-Cortina, R., Zimmer, I., & Rambal, S. (2009) Drought reduced monoterpene emissions from *Quercus ilex* trees: results from a throughfall displacement experiment within a forest ecosystem. *Biogeosciences Discussions*, **6**, 863-893.

Edited books

Bohan, D. A., Dumbrell, A., & **Massol, F.** (2017) *Networks of Invasion: A Synthesis of Concepts*. Vol. 56, *Advances in Ecological Research* series.

Bohan, D. A., Dumbrell, A., & **Massol, F.** (2017) *Networks of Invasion: Empirical Evidence and Case Studies*. Vol. 57, *Advances in Ecological Research* series.

Book chapters

Massol, **F.**, Jabot, F., Manel, S., & Munoz, F. (2015) Etude des réseaux en écologie. In: *Les Réseaux dans les Sociétés Animales* (ed.: Sueur, C.) Editions Matériologiques, 49-98.

Science popularisation & expertise

Massol, F. & Chave, J. (2017) Ecologie fondamentale : les grands défis. In: *Prospectives de l'Institut Ecologie & Environnement du CNRS, Compte-rendu des journées des 22, 23 et 24 Février 2017, Bordeaux*, 7-13.

Invited presentations

Plenary talks at international conferences

2016 Networks of interactions in ecology: Everything you always wanted to know about ecological networks (but were afraid to ask). *Société Française d'Ecologie international conference*, Marseille, France

2014 Individual dispersal, fluxes of nutrients and the stability of ecosystems. *SysBio conference*, Lyon, France

Invited talks at international symposia

2017 The evolution of conditional dispersal under different life cycles. *CIMI conference: Ecology and evolutionary biology, deterministic and stochastic models*, Toulouse, France

2016 Life-history traits and the properties of metaecosystems: a theoretical perspective. *Per Brink thematic symposium, Nordic Oikos conference*, Turku, Finland

2016 Diversification of dispersal rates in variable environments. *Maths-Bio conference day*, Paris, France

2015 Joint evolution of dispersal and the allocation of dispersal cost in heterogeneous landscapes. *Mathematical Models in Ecology and Evolution*, Paris, France

2014 Community dynamics and the evolution of dispersal. *ECMTB meeting*, Göteborg, Sweden

2014 Rescuing metacommunity ecology using random matrix theory. *Journées Modélisation Aléatoire et Statistique*, Toulouse, France

2010 Recasting spatial food web ecology as an ecosystem science. *ESA meeting*, Pittsburgh, USA

Other representative presentations (non-exhaustive lists)

Contributed oral communications

2017 Eco-evolutionary dynamics of colonization rate, patch occupancy and food chain length in model metacommunities. *ESEB meeting*, Groningen, the Netherlands, with V. Calcagno

2017 Effects of cue reliability, environmental uncertainty and the life cycle on the evolution of emigration strategies. *British Ecological Society meeting,* Ghent, Belgium, with F. Débarre

2011 Evolution in temporally variable environments: the importance of life cycles. *Mathematical & Theoretical Ecology conference*, Colchester, UK, with F. Débarre

Talks at workshops

2017 Modelling the dynamics of spatially structured ecosystems. *Ecosystem Theory conference*, Montpellier

2017 Dispersal and the stability of ecosystems. *Hausdorff School: Large random graphs: geometry and applications*, Bonn, Germany

2017 Evolution of dispersal in spatially and temporally variable environments: life cycles and informed dispersal. *FEEDING workshop,* Ghent, Belgium

2012 Metapopulation-based approaches to spatial food webs. *Networks and interactions in agriculture workshop*, Dijon

Seminars

2018 The ecology and evolution of food-chain length in metacommunities. *Sherbrooke, Canada*

2017 Evolution of dispersal in spatially and temporally variable environments. *Louvain-la-Neuve, Belgium*

2016 Evolution of dispersal in spatially and temporally variable environments: life cycles and informed dispersal. *Helsinki, Finland*

2015 Stability, dispersal and ecological networks. *Ecole Polytechnique, Palaiseau, France*

2014 A model for the evolution of dispersal in heterogeneous environments. *Ghent, Belgium*

2013 Adaptive dynamics in a heterogeneous and uncertain world. *Montpellier, France*

2012 The evolution of dispersal: old questions, new answers. *Paris, France*

Julia ASTEGIANO

Born on April 29th 1979 (Córdoba, Argentina), Argentinean nationality.
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ORCID: 0000-0003-0583-7291
http://www.conicet.gov.ar/new_scp/detalle.php?keywords=astegiano&id=47950&datos_academicos=yes
Associated researcher of the Ecology of Interspecific Interactions and Conservation Biology Group, Multidisciplinary Institute of Plant Research (IMBIV), National University of Cordoba - CONICET. CC495, Córdoba, Argentina. CP 5000

RECENT TIMELINE

1998 – 2003 Biologist. National University of Cordoba, Argentine.
2005 – 2010 Ph.D. in Biological Sciences. National University of Cordoba, Argentine.
2011 – 2015 Postdoctoral position, Department of Ecology, Institute of Biosciences, University of São Paulo, Brazil.
2012 – 2013 Postdoctoral position, Dynamic and adaptation of plant populations, Centre d'Ecologie Fonctionnelle et Evolutive (CEFE), France.
2015-2016 Postdoctoral position at the Multidisciplinary Institute of Plant Research.
2016- Associated researcher of CONICET at the Multidisciplinary Institute of Plant Research.

ADVISING AND SUPERVISION

2017- Co-adviser of a PhD. Student at the National University of Córdoba, Argentina. Founding: CONICET.

TEACHING

Argentinean School of Mathematical Biology at the National University of Córdoba, Argentina (edition 2016, 2018; professor).
Ecological Networks, at the University of São Paulo, Brazil (editions 2012 and 2014; assistant professor).
Ecology of the Atlantic Forest, at the University of São Paulo, Brazil (edition 2012; professor).
Inquiring Ecology, at the University of São Paulo, Brazil (editions 2012 and 2014; assistant professor).

FUNDING

2011-2015 Post-doctoral fellowship (FAPESP) on "robustness of plant-pollinator networks and fragmented landscapes: the role of breeding system". PI: Paulo Guimarães and Julia Astegiano.
2012-2013 Post-doctoral fellowship (FAPESP) on "robustness of plant-pollinator networks and fragmented landscapes: the role of breeding system and dispersal strategy on plant species persistence". PI: Paulo Guimarães, Pierre Olivier Cheptou, François Massol, Julia Astegiano.
2017-2019 Project founded by FONCyT on "landscape heterogeneity and resilience of pollination services: basis for the ecological intensification of horticultural crops in Argentina". PI: Julia Astegiano.
2017-2019 Project founded by FONCyT on "ecosystem services provisioned by *Apis mellifera* and native bees and land use". PI: Ramiro Aguilar.
2017-2020 Project founded by CYTED on "the decline of pollinators, pollination

	and plant dispersal ecosystem services in protected areas (SEPODI). PI (Argentina): Lorena Ashworth.
2018	Grant from the Argentinean Minister of Science and Technology (CELFI program) to organize the VIII School on mathematical biology: "Ecosystems dynamics and mathematical models: contributing to the sustainability of agroecological systems". PI: Julia Astegiano, Paula Nieto, Lucas Barberis, Silvina Fenoglio, Gustavo Sibona.

PEER REVIEW IN JOURNALS AND SCIENTIFIC REPPORTS

Oikos; Evolutionary Ecology; Agriculture, Ecosystems and Environment; Advances in Ecological Research; Scientific reports; Ecography; Revista Colombiana de Entomología; Brazilian Journal of Biology.
Evaluator of the IPBES report on Pollination Services.
Manager and recommender of *Peer Community in Ecology* and recommender of *Peer Community in Evolutionary Ecology.*

JURIES

Examiner at 2 PhD and 2 master juries.

REVIEWER OF PROPOSALS FOR PUBLIC FUNDING AGENCIES

CONICET, UdeLaR (National University of Uruguay).

SCIENTIFIC ANIMATION

Co-organizer of conferences (Argentinean Meeting of Ecology, Córdoba 2006; *Argentinean School of Mathematical Biology, Córdoba 2016, 2018*)
Co-organizer of symposia (Bi-national Meeting of Ecology, Argentina-Chile, Puerto Iguazú 2016)
Member of the administrative board of the Argentinean Ecological Society (AsAE) from 2007 to 2008.

PUBLICATIONS (TOP 5 RELATED TO THE PROPOSED RESEARCH)

Astegiano J, Altermatt F, Massol F (2017). Disentangling the co-structure of multilayer interaction networks: degree distribution and module composition in two-layer bipartite networks. *Scientific reports* 7: 15465.

Astegiano J, GuimaraesJr P, Cheptou P-O, Morais Vidal M, Mandai, CY, Ashworth, L, Massol F (2015). Persistence of Plants and Pollinators in the face of Habitat Loss: Insights from Trait-based Metacommunity Models. *Advances in Ecological Research* 53:201-248.

Astegiano J, Massol F, Morais Vidal M, Cheptou P-O, Guimaraes Jr PR (2015). The Robustness of Plant-pollinator Assemblages: Linking Plant Interaction Patterns and Sensitivity to Pollinator Loss. *Plos One* 10(2): e0117243.

Bohan DA, Raybould A, Mulder C, Woodward G, Tamaddoni-Nezhad S, Bluthgen N, Pocock M.J.O., Muggleton S, Evans DM, **Astegiano J**, Massol F, Loeuille N, Petit S, Macfadyen S (2013). Chapter One - Networking Agroecology: Integrating the Diversity of Agroecosystem Interactions. *Advances in Ecological Research* 49: 1-67.

Astegiano J, Funes G, Galetto L (2013). Plant recruitment success and range size:

transition probabilities from ovule to reproductive individuals in two annual sympatric *Ipomoea*. *Acta Oecologica* 48: 76-82.

PUBLICATION SUMMARY

10 articles in peer-reviewed journals; H-index: 6, 119 citations (Google scholar May 29th, 2018). Six articles as senior author, three invited communications at international symposia and workshops and 28 contributions at national and international conferences.

INVITED PRESENTATIONS

2012 **Astegiano J**, Guimarães Jr PR. Plant-pollinator networks and fragmented landscapes: does plants' breeding system matter? 49th Annual Meeting of the Association for Tropical Biology and Conservation (ATBC), Bonito, Brazil. Symposia.

2015 **Astegiano J**. Robustness of plant-pollinator networks and plant syndromes: from real communities to metacommunity models. Atelier Réseaux Trophiques, Agrinaples, Paris, France. Workshop.

2017 **Astegiano J.** Predicting the vulnerability of the pollination benefit in agroecological ecosystems: networks, functional traits and the metacommunity perspective. 1st Meeting of the SEPODI project, Morelia, México. Symposia.

CURRICULUM VITAE (MAY 2018)

DR. RER. NAT. LUTZ BECKS

ADDRESS:	Schillerstrasse 14 24116 Kiel Phone: +49 (0)151 58533983
INSTITUTIONAL ADDRESS:	Max-Planck Institute for Evolutionary Biology Community Dynamics Group August-Thienemann-Str. 2, 24306 Plön Germany Phone: +49 (0)4522 763230, Email: lbecks@evolbio.mpg.de www.evolbio.mpg.de/comdyn
DATE OF BIRTH:	21. April 1978 PLACE OF BIRTH: Jülich
CITIZENSHIP:	Germany

PROFESSIONAL EXPERIENCE

since 10. 2011	Leader of an independent junior research group *(Emmy Noether program and Heisenberg Program, DFG)* at the Max Planck Institute for Evolutionary Biology.
11. 2010 – 10.2011	Postdoctoral Fellow at University of Cologne (Germany) with Prof. Hartmut Arndt (*Volkswagen Foundation Scholarship)*.
10.2009 – 10.2010	Postdoctoral Fellow at University of Toronto (Canada) with Prof. Aneil F. Agrawal *(Volkswagen Foundation Scholarship)*.
08.2006 – 09.2008	Postdoctoral Fellow at Cornell University, Ithaca NY (USA) with Prof. Nelson G. Hairston Jr. and Prof. Stephen P. Ellner.

EDUCATION

03.2004 – 10.2006	Ph.D. in Zoology at the Zoological Institute, University of Cologne (Germany) with Prof. Hartmut Arndt. Thesis title: "Complex population dynamics in microbial systems" (magna cum laude).
02.2002 – 02.2004	Studies and graduation at the University of Cologne (Germany) with a Diploma in Biology. Thesis title: "Chaotic dynamics – a common phenomenon in a two-prey-one-predator system? - Theoretical and experimental analyses of bacteria – protozoan interactions in chemostats" (sehr gut).
08.2001 – 01.2002	Erasmus exchange student at the University of Lund (Sweden).
10.1998 – 07.2001	Studies at the University of Cologne (Germany) in Biology.

PUBLICATIONS

Frickel, J., P. Feulner, E. Karakoc & **L. Becks**. *Population size changes and selection drive patterns of parallel evolution in a host-virus system.* **Nature Communications**, 1706.

Huang, W., Traulsen, A., Werner, B., Hiltunen, T. & **L. Becks**. 2017. *Dynamical trade-offs arise from species coevolution and lower the species diversity.* **Nature Communications**, 2059.

Ehrlich, E., **L. Becks** & U. Gaedke, 2017. *The equalizing trade-off: A key to explain species coexistence and co-occurrence.* **Ecology**, 98, 3188-3198.

Hiltunen, T., Kaitala, V., Laakso, J. & **L. Becks**, 2017. *Evolutionary contribution to coexistence of competitors in microbial food webs*. **Proceedings of the Royal Society B**, 284, 1864.

Frickel, J., L. Theodosiou & **L. Becks**, 2017. *Rapid evolution of hosts begets species diversity at the cost of intraspecific diversity*. **PNAS**, 201701845.

Cairns, J., Frickel, J., Jalasvuori, M., Hiltunen, T. & **L. Becks**, 2017. *Genomic evolution of bacterial populations under co-selection by antibiotics and phage.* **Molecular Ecology**, 26, 1848-1859.

Koch, H. & **L. Becks**, 2017. *The cost of facultative sex in a prey adapting to predation*. **Journal of Evolutionary Biology**, 30, 210-220. (Recommended by Faculty 1000).

Cairns, J., **Becks, L.**, Jalasvuori, M. & T. Hiltunen, 2016. *Sublethal streptomycin concentrations and lytic bacteriophage interactively promote resistance evolution.* **Phil. Trans. R. Soc. B.,** 372, 20160040.

Frickel, J., Sieber, M. & **L. Becks**, 2016. *Eco-evolutionary dynamics in a coevolving host-virus system.* **Ecology Letters,** 19, 450-459.

Haafke, J., Abou Chakra, M. & **L. Becks,** 2016. *Eco-evolutionary dynamics select indirectly for sex in predator populations.* **Evolution,** 70, 641-652.

Koch, H., Jeschke, A. & **L. Becks**, 2015. *Use of ddPCR in experimental evolution studies.* **Methods in Ecology and Evolution**, 7, 340-351.

Hiltunen, T. Ayan, G. & **L. Becks,** 2015. *Environmental fluctuations restrict eco-evolutionary dynamics in predator-prey system*. **Proceedings of the Royal Society B,** 282. doi:/10.1098/rspb.2015.0013.

Becks, L. & Y. Salehi-Alawi, *Using microevolution to explain the macroevolutionary observations for the evolution of sex.* In: Macroevolution: Explanation, Interpretation and Evidence, Volume 2, 279-299 (Springer).

Hiltunen, T. & **L. Becks**, 2014. *Consumer co-evolution as an important component of the eco-evolutionary feedback*. **Nature Communications** 5, 5226, doi:10.1038/ncomms6226.

Koch, H., Frickel, J. Valiadi, M. & **L. Becks**, 2014. *Why rapid, adaptive evolution matters for community dynamics.* **Frontiers in Ecology and Evolution** 2:17. doi: 10.3389/fevo.2014.00017/.

Becks, L. & A.F. Agrawal, 2013. *Higher rates of sex evolve under K-selection.* **Journal of Evolutionary Biology** 26, 900-905.

Becks, L. & H. Arndt, 2013. *Different types of synchrony in chaotic and cyclic communities.* **Nature Communications** 4, doi:10.1038/ncomms2355.

Becks, L. & A.F. Agrawal, 2012. *The evolution of sex is favoured during adaptation to new environments.* **PLoS Biology** 10, 1-11. (Recommended by Faculty 1000).

Becks, L., S.P. Ellner, Jones, L. E. & N.G. Hairston Jr., 2012. *The functional genomics of an eco-evolutionary feedback loop: linking gene expression, trait evolution, and community dynamics.* **Ecology Letters** 15, 492-501. (Recommended by Faculty 1000).

Ellner, S.P. & **L. Becks,** 2011. *Rapid prey evolution in response to two predators*. **Theoretical Ecology** 4,133-152.

Becks, L. & A.F. Agrawal, 2011. *The effect of sex on the mean and variance of fitness in facultatively sexual rotifers*. **Journal of Evolutionary Biology**, 24, 656-664.

Becks, L., S.P. Ellner, Jones, L. E. & N.G. Hairston Jr., 2010. *Reduction of adaptive genetic diversity radically alters eco-evolutionary community dynamics.* **Ecology Letters**, 13, 989-997. (Recommended by Faculty 1000).

Becks, L. & A.F. Agrawal, 2010. *Higher rates of sex evolve in spatially heterogeneous environments*. **Nature,** 468, 89-92. (Recommended by Faculty 1000).

Jones, L. E., **Becks, L.**, Ellner, S.P., Hairston Jr. N.G., Yoshida,T. & G. Fussmann, 2009. *Rapid contemporary evolution and clonal food web dynamics.* **Phil. Trans. R. Soc. B.**, 364,1579-1591.

Becks, L. & H. Arndt, 2008. *Transitions from stable equilibria to chaos, and back, in an experimental food web*. **Ecology**, 89, 3222-3226.

Becks, L., Hilker, F., Malchow. H., Jürgens, K. & H. Arndt, 2005. *Experimental demonstration of chaos in a microbial food web*. **Nature**, 435, 1226-1229. (Recommended by Faculty 1000).

GRANTS AND AWARDS

Total: ***~4,150,000 EUR***

"Host virus coevolution – demography versus selection in the face of multiple stressors" German Research Foundation (DFG) ***334,497 EUR***
"Indirect response to external drivers through trait variation in predator-prey systems" German Research Foundation (DFG), 2018 - 2021 ***382,550 EUR***

Heisenberg Professorship from the German Research Foundation (DFG) at the Christian Albrechts University Kiel, 2017-2020) ***373,000 EUR***

"Eco-evolutionary dynamics in evolving communities" German Research Foundation (DFG), 2017 - 2019 ***233,650 EUR***

"Persistence mechanisms in a tripartite protist — giant virus — virophage system" (Betty and Gordon Moore Foundation) Investigator: L. Becks & M. Fischer (MPI Heidelberg), Jan. 2017 – Dec. 2108 part LB: ***200,000 USD***

"Experimental Evolution and Community Dynamics - EECD 2016" (DFG, International Scientific Events) 2016 ***10,200 EUR***

"Host virus coevolution – demography versus selection" Lead-Agency program" with SNF (P. Feulner EAWAG) as part of the Priority program by the German Research Foundation (DFG) SPP1819: "Rapid Evolution", Jan. 2016 - Dec. 2018 part LB: ***322,328 EUR***

"Evolution in sexual and asexual populations" (Max-Planck Society), 2015 ***80,000 EUR***

"Impact of the trade-offs on eco-evolutionary feedbacks in predator-prey systems" as part of the Priority program (DFG) SPP1704: "Flexibility matters: Interplay between trait diversity and ecological dynamics using aquatic communities as model systems (DynaTrait) "; Principal

Investigator: L. Becks & U. Gaedke (University of Potsdam,) Oct. 2014 - Oct. 2017
part LB: ***187,805 EUR***

"Coevolving communities in spatially structured and heterogeneous environments" (DAAD/Academy of Finland); Principal Investigator: L. Becks & T. Hiltunen, Jan. 2014 - Dez. 2016
part LB: ***10,963 EUR***

"Eco-evolutionary dynamics in complex systems" Emmy Noether Grant (DFG), Oct. 2011 - Oct. 2015 ***1,239,649 EUR***

"The effects of sex on the coevolution and dynamics of a predator-prey system" (Volkswagen Foundation Postdoctoral fellowship), Oct. 2008 - Oct. 2011 ***193,000 EUR***

"Studying complex dynamics: Contemporary rapid evolution in complex ecological communities" (James S. McDonnell Foundation); Principal Investigators: S. P. Ellner, N. G. Hairston Jr., L. Jones, L. Becks (all Cornell University, Ithaca), G. Fussmann (McGill University, Montreal), 2007 - 2011 ***449,459 USD***

Others:
- Shortlisted for W3 position at University Mainz (2nd position; 2015).
- Shortlisted for professorship IST Austria (2016).
- Shortlisted for lecture at Queen Mary University London (2016).
- Young scientist award from the German Limnological Society (DGL), 2005.
- Excellent student scholarship for graduate studies from the University of Cologne (Germany).

INVITED PRESENTATIONS

I have been invited to present my research at internationally established conferences, symposia and advanced schools (**selected for the last 3 years, see below**).

2018	Sex uncovered: the evolutionary biology of reproductive systems, CONFERENCES JACQUES MONOD (France)
2018	Workshop "Experimental approaches to test for coevolution, Paris (France).
2017	Ringberg Symposium - Giant Viruses (Germany); Visions in Science, Berlin (Germany); Workshop "Origin of Biodiversity - Host-pathogen co-evolution in the genomic era" Gothenburg (Sweden).
2016	EVENT, Ghent (Belgium); ETH Zürich (Switzerland); IST Austria (Austria), Workshop "Experimental Evolution: Theory and Current Practices", Paris (France); Workshop "The genomic basis of eco-evolutionary change", Monte Veritas (Switzerland).
2015	European Phycological Congress, London (UK), German Ecological Society, Göttingen (Germany).

SUPERVISION

Bachelor: Niklas Niehus (2015), Annika Friedrichsen (2016).
Master: Noemi Woltermann (2015), Sophia Wagner (2015), Aline Jeschkes (2016),Chris Gundlach (2016), Lasse Harder (2017), Ruben Hermann (ongoing).

PhD: Jens Frickel (defended 26.4.2016), Hanna Koch (defended 9.2016), Gökce Ayan (defended 11. 2017), Loukas Theodosiou (ongoing), Noemi Woltermann (currently on maternal leave). Elena Horas (ongoing).
Postdoc: Martha Valiadi (2.2013 -10.2013), Amrei Binzer (9.2013 - 9.2014), Heike Gruber *Aging and epigenetic modifications* (4.2014 - 1.2016), Vienna Kowallik (1.1.2016 - 31.1.2017), Ana del Arco: (1.4.2017-31.12.2018)

SCIENTIFIC SERVICES

- Faculty of the International Max Planck Research School Evolutionary Biology and member of several thesis committees
- Associate Editor Ecology Letters
- Associate Editor BMC Evolutionary Biology
- Associate Editor Frontiers in Ecology and Evolution

- Organization of the Workshop 'Bridging Theory and Experiments ~Evolving Communities' MPI Plön (2016)
- Organization of Mini-Symposium 'Experimental Evolution and Community Dynamics' University of Helsinki together with Dr. Teppo Hiltunen (2014, 2015) MPI Plön (2016)
- Organization of Symposium "Rapid evolution and population genetics" together with Dr. Teppo Hiltunen at the ESEB meeting in Lisoba (2013)
- Organization of Symposium "Ecology and Evolution of Sex" together with Hanna Koch at the ESEB meeting in Lausanne (2015)
- Organization of Graduate meeting of the German Zoological Society, MPI Plön (2017)
- Co-speaker of DFG priority program DynaTrait
- Co-organization of Kick-off meeting program DynaTrait (2014), annual Meeting (2015, 2016), international meeting (2017)
- Co-organization of Seminar series at the MPI Plön: Meselson Seminar on the Evolution of Sex

- Review for scientific journals: American Naturalist, BMC Biology, BMC Evolutionary Biology, Current Biology, Ecology, Ecology Letters, Evolution, Hydrobiologia, ISME Journal, Journal of Animal Ecology, Journal of Evolutionary Biology, Journal of Phycology, Limnology, Limnology and Oceanography, Molecular Ecology, Molecular Systems Biology, Nature Communications, PNAS, Trends in Ecology and Evolution

- Review or funding agencies: German Research Foundation (DFG), Swiss National Science Foundation (SNSF), Biotechnology and Biological Sciences Research Council (BBSRC), University of Bordeaux, University of Helsinki, FWO (Belgium), ERC.
- Outreach:
    - Darwin Day University of Kiel; Talk for 1200 high school students.
    - Public lectures in 2014, 2015 at the Max Planck Institute for Evolutionary Biology on the evolution of sex.
    - TV interview with public TV station (NDR) explaining the problem of the evolution of sex (2015).

- Scientific representative of the MPI Evolutionary Biology in the Max Planck Society.

CURRICULUM VITAE

1. CONTACT INFORMATION:

MALENA SABATINO, E-mail: malenasabat@gmail.com

Laboratorio de Artrópodos, Universidad Nacional de Mar del Plata.

Funes 3350 (7600) Mar del Plata, Buenos Aires, Argentina.

Tel. +54 223 4752426 int. 450

Scholar Google:https://scholar.google.com.au/citations?user=I6BAhtQAAAAJ&hl=en

ResearchGate: https://www.researchgate.net/profile/Malena_Sabatino

2. CURRENT POSITION:

- Researcher of CONICET (National Council of Scientific Research)

3. UNIVERSITY STUDIES:

- 2003 Graduated degree in Biological Science. Universidad Nacional de Mar del Plata, Buenos Aires, Argentina.
- 2012 Ph.D. in Biological Science. Universidad Nacional de Mar del Plata, Buenos Aires, Argentina.

4. RESEARCH INTERESTS:

My research interest is related to understand how the anthropogenic habitat alteration can affect the diversity and composition of the interactions assemblages, as well as web structure in plant-pollinator communities. Moreover, I am interested in applying the ecological network approach as a tool for developing common strategies for biodiversity conservation and ecological restoration, landscape planning, and sustainable management of natural habitats.

5. GRANTS AND FELLOWSHIPS:

- CONICET Doctoral Grant (2007–2012) Laboratorio de Agroecología, INTA Balcarce, Buenos Aires / Laboratorio Ecotono, INIBIOMA, Río Negro. Argentina.
- FAPESP Postdoctoral Grant (2012) São Paulo State University (UNESP), Río Claro, São Pablo, Brasil.
- CONICET Postdoctoral Grant (2013–2015) Laboratorio Ecotono, UNCOMA / INTA Balcarce, Argentina.

- Argentinean National Agency of Scientific and Technological promotion (founding for young research) PICT-2015-1691 (2016-2018).

6. TEACHING ACTIVITIES:

- 2002–2003 Principles of Geology: Teacher assistant of practical course for graduation students. Universidad Nacional de Mar del Plata, Buenos Aires, Argentina.
- 2006 Extension course: "Plantas aromáticas, extracción de aceites esenciales y usos múltiples". Universidad Nacional de Mar del Plata, Buenos Aires, Argentina.
- 2006–2007 Training extension course of Perito Apicultor. Teacher assistant. Universidad Nacional de Mar del Plata, Buenos Aires, Argentina.
- 2012 Chemistry for the introductory course: Teacher assistant for post-graduate students. Universidad Nacional del Comahue. San Carlos de Bariloche, Río Negro, Argentina.
- 2014 Vascular Plants. Teacher assistant for post-graduate students. Universidad Nacional de Mar del Plata, Buenos Aires, Argentina.
- 2017 Teacher assistant for post-graduate students course: "¿Abejas en crisis?: Una mirada interdisciplinaria sobre la actualidad Argentina y mundial". Universidad Nacional de Mar del Plata, Buenos Aires, Argentina.

7. RESEARCH ADVISOR:

- MSc Student: Agustín Saez, co-supervised with Marcelo Aizen. Universidad Nacional de Mar del Plata, Buenos Aires, Argentina (2009-2010).

8. RESEARCH COLLABORATION AND TRAINING:

- Dr. Jason Tylianakis group from June 2008 to January 2009. School of Biological Sciences, University of Canterbury, New Zealand.
- Dr. Marcelo Aizen group (2007-2012). Universidad Nacional del Comahue, San Carlos de Bariloche, Río Negro, Argentina.
- Dr. Nestor Maceira (2007-2018) EEA INTA Balcarce, Buenos Aires, Argentina.
- Dr. Martín Eguaras (2016-2018) Lab. Artrópodos. Universidad Nacional de Mar del Plata, Buenos Aires, Argentina.

9. PUBLICACTIONS:

- SABATINO M., IURLINA M., EGUARAS M. & R. FRITZ, 2006. Tipificación Botánica y Caracterización de las Mieles del Sudeste de la Provincia de Buenos Aires. *Vida Apícola,* Revista de Apicultura Nº 137. Editores Montagud. Barcelona. España.
- SABATINO M., N. MACEIRA & M. AIZEN, 2010. Direct effects of habitat area on interaction diversity in pollination webs. Ecological Applications, Vol 20(6), pp. 1491–1497.
- HERRERA L. & M.SABATINO, 2010. El valor de los pastizales serranos. Visión Rural. Año XVII Nº 84, pp. 25-27.
- SÁEZ A., SABATINO M. & M. AIZEN, 2011. Interactive Effects of Large- and Small-Scale Sources of Pollinator Service in the Argentine Pampas. PLoS One, (7) 1.
- AIZEN M., SABATINO M. & J. TYLIANAKIS, 2012. Specialization and Rarity Predict Non-Random Loss of Interactions from Mutualist Networks. Science Vol. 335, pp 1486-1489.
- SÁEZ A., SABATINO M. & M.A. AIZEN (2014). La diversidad floral del borde afecta la riqueza y abundancia de visitantes florales nativos en cultivos de girasol. Ecología Austral 24: 94-102.
- CARSTENSEN D. W., SABATINO M., TRØJELSGAARD K. & L. P. C. MORELLATO (2014). Beta diversity of plant-pollinator networks and the turnover of pairwise interactions across space. PLoS ONE 9(11) e112903.
- CHALCOFF, V. R.; CAROLINA L. MORALES; MARCELO A. AIZEN; YAMILA SASAL; ADRIANA E. ROVERE; MALENA SABATINO; CAROLINA QUINTERO; MARIANA TADEY (2014). Interacciones planta-animal, la polinización. En: Raffaele E. de Torres Curth M., Morales C.L. & T. Kitzberger (Eds.), Ecología e historia natural de la Patagonia Andina: un cuarto de siglo de investigación en biogeografía, ecología y conservación. Ciudad Autónoma de Buenos Aires, Fundación de Historia Natural Félix de Azara.
- GILARRANZ L. J., SABATINO M., AIZEN M. A. & J. BASCOMPTE (2015). Hot-spots of mutualistic networks. Journal of Animal Ecology, Vol 84(2), 407-413.
- SABATINO M., N. MACEIRA & A. ROVERE (2015). Germinación de Eryngium regnellii: especie clave para la restauración ecológica de las interacciones planta-polinizador en la Pampa Austral (Buenos Aires, Argentina). Revista Internacional de BOTÁNICA Experimental Phyton Vol 84: 435-443.

- AIZEN, M. A., GLEISER, G., SABATINO, M., GILARRANZ, L. J., BASCOMPTE, J., & VERDÚ, M. (2016). The phylogenetic structure of plant–pollinator networks increases with habitat size and isolation. *Ecology letters*, *19*(1), 29-36.
- CARSTENSEN D. W., SABATINO M. & L. P. C. MORELLATO (2016). Modularity, pollination systems, and interaction turnover in plant-pollinator networks across space. Ecology 97 (5), 1298-1306.
- SABATINO M. & N. MACEIRA (2016). El rol de las sierras en la conservación de la biodiversidad pampeana. Visión Rural. Año XXIII Nº 112, pp. 46-48.
- HERRERA L., M. SABATINO, A. GASTÓN, S. SAURA (2017). Grassland connectivity explains entomophilous plant species assemblages in an agricultural landscape of the Pampa Region, Argentina. Austral Ecology 42: 486-496.
- HERRERA L., M. SABATINO, F. R. JAIMES & S. SAURA (2017). Landscape connectivity and the role of small habitat patches as stepping stones: An assessment of the grassland biome in South America. Biodiversity and Conservation 26: 3465.
- HERRERA L., F. R. JAIMES, M. SABATINO & S. L. POGGIO (2017). Una propuesta para valorar el estado de conservación de los bordes de caminos rurales en el sudeste bonaerense. Ecología Austral 27:404-414.
- SABATINO M., FARINA J. & MACEIRA N. (2017). Flores de las sierras de Tandilia: Guía para el reconocimiento de las plantas y sus visitantes florales. INTA Ediciones, Colección Recursos.

10. CONGRESS PARTICIPATION (Last 5 years):

- CARSTENSEN D.W., SABATINO M., MORELLATO L.P.C. Spatial turnover of pairwise plant-pollinator interactions. ATBS 2013. San José, Costa Rica.
- HERRERA L.P., JAIMES F., SABATINO M., POGGIO S. Una nota al margen: una propuesta para valorar el estado de conservación de los bordes de caminos en el Sudeste Bonaerense. XXVI Congreso Argentino de Ecología. Libro de resúmenes RAE, Noviembre 2014. Rio Negro, Argentina.
- JAIMES F., SABATINO M., HERRERA L. Mapeo de pastizales remanentes del Sistema de Tandilia (Buenos Aires): Un diagnóstico de la situación actual. XXVI Congreso Argentino de Ecología. Libro de resúmenes RAE, Noviembre 2014. Rio Negro, Argentina.
- ROVERE A.E., SABATINO M. Restauración ecológica de las interacciones planta-polinizador en la matriz agrícola Pampeana. IV Congreso Iberoamericano y del

Caribe de Restauración Ecológica. Libro de resúmenes SIACRE, Abril 2015. Buenos Aires, Argentina.

- ROVERE A.E., SABATINO M., BRAN D., FARINA J. Especies claves para la restauración de las interacciones planta-polinizador en un área degradada de la Estepa Patagónica. III° Taller regional sobre rehabilitación y restauración de la Diagonal Árida de la Argentina. Libro de resúmenes, Octubre 2016. Puerto Madryn, Chubut, Patagonia Argentina.
- SABATINO M., FARINA JUAN, MACEIRA N. Flores de las sierras de de Tandilia: Guía para el reconocimiento de las plantas y sus visitantes florales. Comunicación en la 1º Reunión Científica del Programa Nacional de Recursos Naturales, Gestión Ambiental y Ecorregiones: aportes a la agroecología desde la biodiversidad, la gestión ambiental, el estudio del clima y el ordenamiento territorial. Libro de resúmenes, Agosto 2016, Buenos Aires, Argentina.

Matías Arim

Current affiliation: Department of Ecology and Evolution, University of República (Universidad de la República), Maldonado, Uruguay, VA 20000. Ph: (598) 98 119807

Research focus: the interaction between landscape structure, energy, and biodiversity, as determinants of metacommunity organization and function. I particularly attempt to connect theoretical and empirical studies.

A. Education:

The University of República, Uruguay	1999	B. Sc.	Biology
The University of República, Uruguay	2001	M.Sc.	Ecology
P. Catholic University of Chile, Chile	2005	PhD	Ecology
Center for Advanced Studies in Ecology and Biodiversity, Chile	2006	Postdoc	Ecology

B. Appointments:

2015-Present	Full Professor, Program for the Development of Basic Science, Uruguay
2005-2009	Assistant Professor, University of República, Uruguay
2009-Present	Associate Professor, University of República, Uruguay
2007-2011	CoPI, Center for Advanced Studies in Ecology and Biodiversity, CASEB, Chile (P. Catholic University of Chile).
2014-2018	Member of the Global Young Academy

C. Publications *Highlighted from 4 books chapters and 50 papers ISI*

1. **Arim, M**. & O. Barbosa. 2002. **Humped pattern of diversity: fact or artifact?** *Science.* 297:1763. http://www.sciencemag.org/cgi/reprint/297/5588/1763a.pdf
2. **Arim, M.** & P.A. Marquet. 2004. **Intraguild Predation: a significant and widespread interaction**. *Ecology Letters* 7: 557-564.
3. **Arim M.** & F. M. Jaksic. 2005. **Productivity and food web structure: association between productivity and link richness among top predators**. *Journal of Animal Ecology* 74: 31-40.
4. [1]**Arim, M**, S.R. Abades, P.E. Neill, M. Lima & P.A. Marquet. 2006. **Spread dynamics of invasive species**. *Proceeding of the National Academy of Science USA*. *PNAS* 103: 374-378.
5. **Arim, M**, Pablo A. Marquet & F.M. Jaksic. 2007. **On the relationship between productivity and food chain length at different ecological levels**. *The American Naturalist* 169: 62-72.
6. **Arim, M**, F. Bozinovic & Pablo A. Marquet. 2007. **On the relationship between trophic position, body mass and temperature: Reformulating the energy limitation hypothesi**s. *OIKOS*. 116: 1524-1530.

1 Articles recommended in "Faculty of 1000".

7. Lafferty, K.D.; Allesina S.; **M. arim**; Briggs, C.J.; De Leo, G.; Dobson, A.P.; Dunne, J.A.; Johnson, P.T.J.; Kuris, A.M.; Marcogliese, D.J.; Martinez, N.D.; J. Memmott, J.; P.A. Marquet; J.P. Mclaughlin; E.A. Mordecai; M. Pascual; R. Poulin; D.W. Thieltges, D.W. 2008. **Parasites in food webs: the ultimate missing links**. *Ecology Letters* 11: 533-546.
8. Ramos-Jiliberto, R., A. Albornoz, F. Valdovinos, C. Smith-Ramírez, **M. Arim**, J. Armesto, P.A. Marquet. 2009**. Mutualistic networks, pollination web, degree distribution, nestedness, network dynamics**. *Oecologia.* 160: 697-706.
9. **Arim, M.**, S.R. Abades, G. Laufer, M. Loureiro & P.A. Marquet. 2010 **Food web structure and body size: trophic position and resource acquisition**. Oikos 119:147-153.
10. Toranza, C. & **M. Arim**. 2010. Cross-taxon congruence and environmental conditions. BMC Ecology 10:18.
11. Carranza,A. Defeo,O. & **Arim, M. 2**011. **Taxonomic relatedness and spatial structure of a shelf benthic gastropod assemblage.** *Diversity and Distribution* 17: 25-34.
12. **Arim, M**., M. Berazategui, J. M. Barreneche, L. Ziegler, M. Zaruki, and S. R. Abades. 2011. **Determinants of density–body size scaling within food webs and tools for their detection.** *Advances in Ecological Research* **45**:1-40.
13. Borthagaray, A.I., **Arim, M.** and Marquet, P.M. 2012. **Connecting landscape structure and patterns in body size distribution**. *Oikos* DOI: 10.1111/j.1600-0706.2011.19548.x).
14. Garay-Narvaez, L., **Arim, M**., Flores, J.D., Ramos-Jilberto, R. 2013. **The more polluted the environment, the more important biodiversity is for food web stability**. Oikos, 122: 1247-1253.
15. *Ziegler L., Berazategui M. &* ***Arim M.*** *2014* **Discontinuities and alternative scalings in the density–mass relationship of anuran larvae**. *Hydrobiologia. 723:123-129. doi: 10.1007/s10750-013- 1553-2.*
16. Borthagaray, A.I., Barreneche, J.M. Abades, S.R., & **Arim, M**. 2014 **Modularity along organism dispersal gradients challenges a prevailing view of abrupt transitions in animal landscape perception**. *Ecography*. 37(6):564-571. DOI: 10.1111/j.1600-0587.2013.00366.x
17. Garay-Narvaez, L., Flores, J.D., **Arim, M**., Ramos-Jilberto, R. 2014. **Food web modularity and biodiversity promote species persistence in polluted environments**. Oikos, 123: 583-588. DOI: 10.1111/j.1600-0706.2013.00764.x
18. Borthagaray, A.I., **Arim, M.**, Marquet, P.A. 2014. **Inferring species roles in metacommunity structure from species co-occurrence networks**. *Proceedings of the Royal Society B*. 281:. DOI: http://dx.doi.org/10.1098/rspb.2014.1425
19. Segura, A., Calliari, D. Kruk, C., Fort, H.,Izaguirre, I., Saad, J., **Arim, M**. 2015 **Metabolic dependence of phytoplankton species richness**. Global Ecology and Biogeography 24: 472-482. DOI: 10.1111/geb.12258
20. Borthagaray, A.I., Berazategui, M., & **Arim, M**. 2015. **Disentangling the effects of local and regional processes on biodiversity patterns through taxon-contingent metacommunity network analysis**. *Oikos*. 124: 1383-1390. DOI: 10.1111/oik.01317
21. Borthagaray, A.I., Pinelli, V., Berazategui, M., Rodríguez-Tricott, L. and **M. Arim.** 2015. **Effects of metacommunity network on local communities structure: from theoretical predictions to empirical evaluations**. En Aquatic Functional Biodiversity A. Belgrano, G. Woodward and U. Jacob (Eds). Elsevier
22. **Arim, M.**, Borthagaray, A.I., Giacomini, H.C. 2016. **Energetic constrains to food chain lenght in a metacommunity framework**. Candaina Journal of Fisheries and Aquatic Sciences. 73(4): 685-692. DOI: 10.1139/cjfas-2015-0156
23. Segura, A. M., Fariña, R. A., & **Arim, M.** 2016. **Exceptional body sizes but typical**

trophic structure in a Pleistocene food web. *Biology Letters*, *12*(5), 20160228
24. Segura, A. M., Calliari, D., Lan, B. L., Fort, H., Widdicombe, C. E., Harmer, R., & **Arim, M.** 2017. **Community fluctuations and local extinction in a planktonic food web**. *Ecology Letters*, *20*(4), 471-476.
25. Canavero, A., **Arim, M**., Pérez, F., Jaksic, F. M., & Marquet, P. A. (2018). **A metabolic view of amphibian local community structure: the role of activation energy**. *Ecography.41: 388-400.*
26. Borthagaray, A.I., Soutullo, A., Carranza, A. **Arim, M**. 2018. A modularity-based approach for identifying biodiversity management units. Revista Chilena de Historia Natural. 91: 2.

D. Synergistic Activities:

D.1. Reviewer for Journals:

1. Acta Oecologica
2. Aquatic Ecology
3. Biological Invasions
4. Biological Research
5. BMC Ecology
6. Boletín de la Sociedad Zoológica del Uruguay
7. Canadian Journal of Fisheries and Aquatic Sciences
8. Canadian J. of Zoology
9. Climate Research
10. Conservation Biology
11. Diversity and Distribution
12. **Ecography**
13. Ecological Entomology
14. Ecological Modelling
15. **Ecology**
16. **Ecology Letters**
17. Ecology and Evolution
18. Ecosistemas
19. Evolutionary Ecology Research
20. Frontiers in Ecology
21. **Functional Ecology**
22. F1000Research
23. Gayana
24. **Global Ecology and Biography**
25. Hydrobiologia
26. Ibis
27. **Journal of Animal Ecology**
28. Journal of Arid Environments
29. **Journal of Biogeography**
30. **Journal of Ecology**
31. Journal of Mammalogy
32. Journal of Zoology
33. **Marine Ecology Progress Series**
34. Marine Biodiversity
35. Neotropical Ichthyology
36. **Oecologia**
37. **Oikos**
38. Revista Chilena de Historia Natural
39. **Theoretical Ecology**
40. Wildlife Research

D.2. Reviewer of proposals

National Science Organization of Argentina, Austria, Chile, Colombia, Panamá, Uruguay, and for the European Research Council.

E. Advisor

Advisor of 16 undergraduate, 16 master, and 8 PhD students in programs of Argentine, Chile, Spain, and Uruguay. Now advisor of 3 Master, and 4 PhD students.

F. Principal Investigator in Grants *(highlighted from a total of 10)*

Arim, M. 2008-2018. Programs for University Development. PI of the Program "Development of Research and Human Resources on Biodiversity"
Arim, M. **2013-2015.** "Metabolism, randomness, scale, and entropy: empirical evaluation of new theories in community Ecology" (Fondo Clemente Estable 2011, Metabolismo, azar, escalas y entropía: evaluación empírica de nuevas teorías en ecología de comunidades).
Arim, M. **2016-2018**. "Effect of metacommunity network on local communities: from theoretical predictions to empirical evaluations" (Fondo Clemente Estable 2014, Efecto de la red metacomunitaria en las comunidades locales: desde las predicciones teóricas a la evaluación empírica). 2016-2017.

G. Responsible of undergraduate and graduate courses

Theory and methods in metacommunity ecology. Course directed to graduate students, organized with Dr. Ana Borthagaray in Argentina and Uruguay (since 2017).

Community Ecology. Course directed to graduate students (since 2012).

Ecology. Course directed to undergraduate students (since 2013).

Theoretical Ecology. Course directed to undergraduate students (since 2013).

F. Presentations

1. **2018. Arim M.**, Pinelli V., Rodríguez L., Ortiz E., Illarze M., Fagúndez C., Borthagaray A. **Ponds size and isolation determining the relative role of niche versus neutral assembly of plants communities**. Nordic Society OIKOS, Trondheim, Norway.
2. **2017. Arim, M**. **Gastropods in food webs from general predictions to idiosyncratic patterns**. X Latin-American Congress of Malacology, Maldonado, Uruguay. Plenary Conference.
3. **2016. Arim, M. Metacommunities and Ecological Theory Integration.** Network for Ecological Theory Integration (NETI), Satan Fe Institute, USA.
4. **2015. Arim, M. The importance of take a lock outside the pond to understand pond ecology**. F. de Ciencias, Universidad de la República, Uruguay. II Simposio de Peces Anuales. Plenary Conference.
5. **2015. Arim, M. Networks impacting networks: the interrelationship between metacommunities and food webs**. XXII Reunión Anual de la Sociedad de Ecología de Chile. Plenary Conference*.*

Ruben Hüttel Heleno, Ph.D

Full name:	Ruben Hüttel Heleno
Current position:	Auxiliary researcher (Investigador FCT)
Institution:	Centre for Functional Ecology, Department of Life Sciences, University of Coimbra
Hopepage	http://cfe.uc.pt/#/profile/members/14
Google scholar	https://scholar.google.co.uk/citations?user=**PR5Wrr4AAAAJ**&hl=pt-PT
ResearcherID	http://www.researcherid.com/rid/**A-5778-2011**
citations:	Web of Science= 591; Google Scholar= 101
h-index:	Web of Science= 15 ; Google Scholar= 19

Higher education

- **2005-2009** PhD in Community Ecology: "The impact of alien plants on native biota: a food Web approach", School of Biological Sciences, University of Bristol, UK. ("Doutoramento" registered at the University of Coimbra in July 2013). No corrections.
- **1999-2004** Biology Degree, specialization in Animal Ecology, (average grade 16/20), University of Coimbra, Portugal.

Selected publications ISI (5 most relevant highlighted)

51 J Costa, J Ramos, L Silva, S Timóteo, P Andrade, P Araújo, C Carneiro, E Correia, P Cortez, M Felgueiras, C Godinho, R Lopes, C Matos, A Norte, P Pereira, A Rosa, R Heleno *in press* Rewiring of experimentally disturbed seed dispersal networks might lead to unexpected network configurations. **Basic and Applied Ecology** (I.F.=2.1; Q1)

50 Correia M, R Heleno, P Vargas, S Rodriguez-Echeverría *in press* Should I stay or should I go? Mycorrhizal plants are more likely to invest in long-distance seed dispersal than non-mycorrhizal plants. **Ecology Letters** doi:10.1111/ele.12936 (I.F.=9.45; Q1)

49 Olesen J, C Damgaard, F Fuster, R Heleno, M Nogales, B Rumeu, K Trøjelsgaard, P Vargas, A Traveset 2018 Disclosing the double mutualist role of birds on Galápagos. **Scientific Reports** 8:57 doi:10.1038/s41598-017-17592-8 (I.F.=4.85; Q1)

48 Timóteo S, M Correia, S Rodríguez-Echeverría, H Freitas, R Heleno 2018 Multilayer networks reveal the spatial structure of seed-dispersal interactions across the Great Rift landscapes. **Nature Communications** 9:140 doi:10.1038/s41467-017-02658-y (I.F.=12.12; Q1)

47 Vargas P, M Fernández-Mazuecos, R Heleno 2018 Phylogenetic evidence for Miocene origin of Mediterranean lineages: species diversity, reproductive traits and geography. **Plant Biology** 20(S1):157-165 doi:10.1111/plb.12626 (I.F.=2.10; Q1)

45 Arjona Y, M Nogales, R Heleno, P Vargas 2018 Long-distance dispersal syndromes matter: diaspore-trait effect on shaping plant distribution across the Canary Islands. **Ecography** doi:10.1111/ecog.02624 (I.F.=5.41; Q1)

44 Rumeu B, M Devoto, A Traveset, J Olesen, P Vargas, M Nogales, R Heleno 2017 Predicting the consequences of disperser extinction: richness matters the most when abundance is low. **Functional Ecology** 31(10): 1910-1920, doi:10.1111/1365-2435.12897 (I.F.=5.1; Q1)

43 Nogales M, A González-Castro, B Rumeu, A Traveset, P Vargas, P Jaramillo, J Olesen, R Heleno 2017 Contribution by vertebrates to seed dispersal effectiveness in the Galápagos islands: a community-wide approach. **Ecology** 98(8) 2049–2058, doi:10.1002/ecy.1816 (I.F.=4.7; Q1)

42 Patiño J, R Whittaker, P Borges, J Fernández-Palacios, C Ah-Peng, M Araújo, S Ávila, P Cardoso, J Cornuault, E de Boer, L de Nascimento, A Gil, A González-Castro, D Gruner, R Heleno, J Hortal, J Illera, C Kaiser-Bunbury, T Matthews, A Papadopoulou, N Pettorelli, J Price, A Santos, M Steinbauer, K Triantis, L Valente, P Vargas, P Weigelt, B Emerson, 2017. A roadmap for island biology: 50 fundamental questions after 50 years of The Theory of Island Biogeography. **Journal of Biogeography** 44(1):963-983, doi:10.1111/jbi.12986 (I.F.=4.00; Q1)

41 Heleno R 2017 Birds make the World go round. **Trends in Ecology and Evolution** 32(3):154-155, doi:10.1016/j.tree.2017.01.001 (I.F.=16.74; Q1)

40 López-Núñez F, R Heleno, S Ribeiro, H Marchante, E Marchante 2017. Four-trophic level food webs reveal the cascading impacts of an invasive plant targeted for biocontrol. **Ecology** 98(3):782-793, doi:10.1002/ecy.1701 (I.F.=4.7; Q1)

39 Guzmán B, R Heleno, M Nogales, W Simbaña, A Traveset, P Vargas 2017 Evolutionary history of the endangered shrub snapdragon (*Galvezia leucantha*) of the Galápagos Islands. **Diversity and Distributions** 23(3):247-260, doi:10.1111/ddi.12521 (I.F.=4.6; Q1)

38 da Silva L, J Ramos, A Coutinho, P Tenreiro, R Heleno 2017 Flower visitation by European birds offers the first evidence of interaction release in continents. **Journal of Biogeography** 44(3):687-695, doi:10.1111/jbi.12915 (I.F.=4.00; Q1)

37 Rodríguez-Echeverría S, H Teixeira, M Correia, S Timóteo, R Heleno, M Öpik, M Moora 2017 Arbuscular mycorrhizal fungi communities from tropical Africa reveal strong ecological structure. **New Phytologist** 213(1):380–390, doi:10.1111/nph.14122 (I.F.=7.21; Q1)

36 Correia M, S Timóteo, S Rodríguez-Echeverría, A Mazars-Simón, R Heleno 2017 The refaunation of Gorongosa National Park, Mozambique and the reinstatement of the seed dispersal function. **Conservation Biology** 31(1):76-85, doi:10.1111/cobi.12782 (I.F.=4.27; Q1)

34 da Silva L, A Coutinho, R Heleno, P Tenreiro, J Ramos 2016 Dispersal of fungi spores by non-specialized flower-visiting birds. **Journal of Avian Biology** 47(3):438–442, doi:10.1111/jav.00806 (I.F.=1.97; Q1)

33 Traveset A, M Nogales, P Vargas, J Olesen, P Jaramillo, R Heleno 2016 The Galápagos land iguana (*Conolophus subcristatus*) as a seed disperser. **Integrative Zoology** 11(3):207-213, doi:10.1111/1749-4877.12187 (I.F.=1.9; Q1)

32 Costa J, L da Silva, J Ramos, R Heleno 2016 Sampling completeness in seed dispersal networks: When enough is enough. **Basic and Applied Ecology** 17(2):155-164, doi:10.1016/j.baae.2015.09.008 (I.F.=2.4; Q1)

31 Traveset A, C Tur, K Trøjelsgaard, R Heleno, R Castro-Urgal, J Olesen 2016 Global patterns of mainland and insular pollination networks. **Global Ecology and Biogeography** 25(7):880-890, doi:10.1111/geb.12362 (I.F.=7.26; Q1)

30 Nogales M, R Heleno, B Rumeu, A González-Castro, A Traveset, P Vargas, J Olesen 2016 Seed-dispersal networks on the Canaries and the Galápagos archipelagos: Interaction modules as biogeographical entities. **Global Ecology and Biogeography** 25(7):912-922, doi:10.1111/geb.12315 (I.F.=7.26; Q1)

29 Esteves C, J Costa, P Vargas, H Freitas, R Heleno 2015 On the limited potential of Azorean fleshy fruits for oceanic dispersal. **PLoS One** 10(10):e0138882, doi:10.1371/journal.pone.0138882 (I.F.=3.2; Q1)

28 Traveset A, S Chamorro, J Olesen, R Heleno 2015 Space, Time and Aliens: charting the dynamic structure of Galápagos pollination networks. **AoB Plants** 7:plv068, doi:10.1093/aobpla/plv068 (I.F.=2.27; Q2)

27 Vargas P, Y Arjona, M Nogales, R Heleno 2015 Long-distance dispersal to oceanic islands: success of plants with multiple diaspore specializations. **AoB Plants** 7:plv073, doi:10.1093/aobpla/plv073 (I.F.=2.27; Q2)

26 Traveset A, J Olesen, M Nogales, P Vargas, P Jaramillo, E Antolín, M Trigo, R Heleno 2015 Bird-flower visitation networks in the Galápagos unveil a widespread interaction release. **Nature Communications** 6:6376, doi:10.1038/ncomms7376 (I.F.=10.74; Q1)

24 Heleno R, P Vargas 2015 How do islands become green? **Global Ecology and Biogeography** 24(5):518-526, doi:10.1111/geb.12273 (I.F.=7.26; Q1)

23 da Silva L, J Ramos, J Olesen, A Traveset, R Heleno 2014 Flower visitation by birds in Europe **OIKOS** 123:1377-1383, doi:10.1111/oik.01347 (I.F.=3.56; Q1)

19 Heleno R, C Garcia, P Jordano, A Traveset, JM Gómez, N Blüthgen, J Memmott, M Moora, J Cerdeira, S Rodríguez-Echeverría, H Freitas, J Olesen 2014 Ecological networks: Delving into the architecture of biodiversity. **Biology Letters** 10(1):20131000, doi:10.1098/rsbl.2013.1000 (I.F.=3.53; Q1)

18 Costa J, J Ramos, L da Silva, S Timóteo, P Araújo, M Felgueiras, A Rosa, C Matos, P Encarnação, P Tenreiro, R Heleno 2014 Endozoochory largely outweighs epizoochory in migrating passerines. **Journal of Avian Biology** 45:59-64, doi:10.1111/j.1600-048X.2013.00271.x (I.F.=2.35; Q1)

17 Cruz J, J Ramos, L da Silva, P Tenreiro, R Heleno 2013 Seed dispersal networks in an urban novel ecosystem. **European Journal of Forest Research** 132(5):887-897, doi:10.1007/s10342-013-0722-1 (I.F.=1.8; Q1)

16 Traveset A, R Heleno, S Chamorro, P Vargas, C McMullen, R Castro-Urgal, M Nogales, H Herrera, J Olesen 2013 Invaders of pollination networks in the Galápagos Islands: emergence of novel communities. **Proceedings of the Royal Society - B** 280(1758):2012-3040, doi:10.1098/rspb.2012.3040 (I.F.=5.68; Q1)

15 Heleno R, J Olesen, M Nogales, P Vargas, A Traveset 2013 Seed dispersal networks in the Galápagos and the consequences of alien plant invasions. **Proceedings of the Royal Society - B** 280(1750):2012-2112, doi:10.1098/rspb.2012.2112 (I.F.= 5.68; Q1)

14 Heleno R, J Ramos, J Memmott 2013 Integration of exotic seeds into an Azorean seed dispersal network. **Biological Invasions** 15(5):1143-1154, doi:10.1007/s10530-012-0357-z (I.F.=2.8; Q1)

12 Nogales M, R Heleno, A Traveset, P Vargas 2012 Evidence for overlooked mechanisms of long-distance seed dispersal to and between oceanic islands. **New Phytologist** 194(2):313-317, doi:10.1111/j.1469-8137.2011.04051.x (I.F.=6.37; Q1)

11 Heleno R, M Devoto, M Pocock 2012 Connectance of ecological networks and conservation value. **Ecological Indicators** 14(1):7-10, doi:10.1016/j.ecolind.2011.06.032 (I.F.=3.67; Q1)

10 Vargas P, R Heleno, M Nogales, A Traveset 2012 Colonization of the Galápagos Islands by plants with no specific syndromes for long-distance dispersal: a new perspective. **Ecography** 35(1):33-43, doi:10.1111/j.1600-0587.2011.06980.x (I.F.=4.02; Q1)

9 Heleno R, S Blake, P Jaramillo, A Traveset, P Vargas, M Nogales 2011 Frugivory and seed dispersal in the Galápagos: What is the state of the art? **Integrative Zoology** 6 (2): 110-119, doi:10.1111/j.1749-4877.2011.00236.x (I.F.=1.42; Q1)

8 Heleno R, G Ross, A Everard, J Memmott, J Ramos 2011 On the role of avian seed predators as seed dispersers. **Ibis** 153 (1): 199-203, doi:10.1111/j.1474-919X.2010.01088.x (I.F.=1.92; Q1)

5 Ceia R, H Sampaio, S Parejo, R Heleno, M Arosa, J Ramos, G Hilton 2010 Throwing the baby out with the bathwater: does laurel forest restoration remove a critical winter food supply for the critically endangered Azores bullfinch? **Biological Invasions** 13(1):93-104, doi:10.1007/s10530-010-9792-x (I.F.=2.8; Q1)

4 Heleno R, I Lacerda, J Ramos, J Memmott 2010 Evaluation of restoration effectiveness: community response to the removal of alien plants. **Ecological Applications** 20(5):1191-1203, doi:10.1890/09-1384.1 (I.F.=4.21; Q1)

1 Heleno R, R Ceia, J Ramos, J Memmott 2009 The effect of alien plants on insect abundance and biomass: a food web approach. **Conservation Biology** 23(2):410-419, doi:10.1111/j.1523-1739.2008.01129.x (I.F.=5.16;Q1)

Books and chapters

7 Escribano-Avila G, C Lara-Romero, R Heleno, A Traveset 2018 Tropical Seed dispersal Networks: emerging patterns, biases and keystone species traits. *in* Ecological Networks in the Tropics. W Dattilo and V Rico-Gray (eds), Springer, 93-110 ISBN-13:9783319682273.

6 Traveset A, R Heleno, M Nogales 2014 The ecology of seed dispersal. *in* Seeds: The ecology of regeneration in plant communities. R Gallagher (ed), Oxon, UK, CABI International, 62-93.

5 Gardener M, M Trueman, C Buddenhagen, R Atkinson, R Heleno, H Jäger, A Tye 2013 A pragmatic approach to management of plant invasions in Galapagos. *In* Plant Invasions in Protected Areas: Patterns, Problems and Challenges, Invading Nature - Springer Series in Invasion Ecology 7. C Llewellyn, D Richardson, P Pyšek, P Genovesi (eds) Dordrecht, Springer Science+Business Media, 349-374.

4 Jaramillo P, R Heleno 2012 Seeds of the Galápagos Islands. Puerto Ayora, Galapagos, Charles Darwin Foundation.

3 Borges P et al. 2010 List of Arthropods (Arthropoda) of the Azores. *in* P Borges et al. (ed). A list of the terrestrial and marine biota from the Azores. Princípia Editora, Cascais.

2 Several authors 2008 Novo guia das Aves Nidificantes de Portugal, edited by the Sociedade Portuguesa para o Estudo das Aves and Instituto de Conservação da Natureza e das Florestas

1 Memmott J, R Gibson, L Carvalheiro, K Henson, R Heleno, M Lopezaraiza, S Pearce 2007 The conservation of ecological interactions. *in* A Stewart, T New, O Lewis (eds). Insect Conservation Biology. The Royal Entomological Society, London.

Selected oral presentation (invited plenary/keynote talks highlighted)

53 Heleno R 2017 On the road to a novel natural world: How alien species change the structure and function of ecological networks. 14th EMAPI - Ecology and Management of Alien Plant Invasions, Lisbon, Portugal – **INVITED PLENARY KEYNOTE**

51 Heleno R 2017 Non-standard endozoochory and epizoochory by small passerines in the Azores, Galapagos and mainland Portugal. Workshop "Beyond frugivory and scatterhoarding" EBD-CSIC, Sevilla, Spain **– INVITED**

49 Heleno R 2016 Macroecological patterns of species interaction networks: community assembly, "evolution" and disassembly, Trends in Biodiversity and Evolution – TiBE2016, CIBIO – Universidade do Porto, Oporto, Portugal – **INVITED KEYNOTE**

48 Heleno R, A Coelho, F Mendes, R de Lima 2016 The impact of introduced mammals on the seed-dispersal network of São Tomé island, Island Ecology Symposium, University of Seychelles, Mahé, Seychelles – **INVITED**

43 Heleno R 2016 The role of birds on ecosystem functions: a network approach, VI Iberian Ornithological Congress – SPEA & SEO, Vila Real, Portugal – **INVITED PLENARY KEYNOTE**

37 Heleno R, Y Arjona, M Nogales, P Vargas 2015 Do dispersal syndromes matter? Inter-island plant dispersal across the Azores, Canaries, Galápagos and Cape Verde, European Ecological Federation meeting, Rome, Italy **– INVITED**

33 Heleno R 2015 Interaction modules: disentangling the complexity of biological communities, 4th Iberian Ecological Congress, Coimbra, Portugal – **INVITED PLENARY KEYNOTE**

24 Heleno R 2014 Birds in species-interaction networks: where do we stand and where can we go from here?, British Ornithological Union one-day meeting: Birds in the entangled bank, Peterborough, UK – **INVITED KEYNOTE**

5 Heleno R 2008 Seed Dispersal Networks: Are Predators That Irrelevant for Seed Dispersal?, EGI Student Conference, University of Oxford, **Oxford**, UK – **HIGH HONOUR MENTION FOR BEST TALK**

Projects (raised >2.2M€ in funding)

- **International funding as principal investigator: 2013-2016** "**SEEDS** - Closing the seed dispersal loop: How does seed dispersal affect plant population structure at the global, regional and local scales?" Centre for Functional Ecology, University of Coimbra. Marie Curie Career Integration Grant, European Union (FP7-PEOPLE-2012-CIG-321794) 100.000€.
- **International funding as team member: 2018-2020** "**ISLET-FOODWEBS** - Effects of Global Change on small island trophic meta-networks" PI: Anna Traveset, Institut Mediterrani d'Estudis Avançats (CSIC-UIB), Spain. Spannish Ministerio de Economia y Competitividad (PE2017: CGL2017-88122-P) 196.000€.
- **International funding as team member: 2017-2020** "Spatio-temporal responses of ecosystem functions to ecological restoration" PI: Christopher Kaiser-Bunbury, Darmstadt Technical University; Deutsche Gesellschaft für Finanzwirtschaft (DGF 2017 KA3349/2-2) 436.300€.
- **International funding as team member: 2016-2018** "**COLONIZATION** - Características dispersivas: reconstrucción de la colonización de especies vegetales entre islas oceánicas" PI: Pablo Vargas, Real Jardín Botánico de Madrid (CSIC-RJB). Spannish Ministerio de Economia y Competitividad (PN2014: SPID201500X067883IV0) 168.205€.

- **International funding as team member: 2015-2017** "**VERMUTIS** - Importance of single and double plant-vertebrate mutualisms on islands: Double benefits, double risks?" PI: Anna Traveset, Institut Mediterrani d'Estudis Avançats (CSIC-UIB), Spain. Spannish Ministerio de Economia y Competitividad (PE2013: CGL2013-44386-P) 183.000€.
- **International funding as team member: 2013-2016** "**OCEIS** -The colonization history in the Galapagos islands by plants: floristic, diaspore and phylogenetic analysis". PI: Pablo Vargas, Real Jardín Botánico de Madrid (CSIC-RJB). Spannish Ministerio de Economia y Competitividad (PN2012: CGL2012-38624-C02-01) 140.000€.
- **International funding as team member: 2010-2012** "**REDGAL** - Redes mutualistas en las islas Galápagos: Impactos directos e indirectos de especies invasoras sobre plantas amenazadas" PI: Anna Traveset, Institut Mediterrani d'Estudis Avançats (CSIC-UIB), Spain. Fundación BBVA 199.166€.
- **International funding as team member: 2016** "Alien hitchhiker's: The role of avian seed dispersal to São Tomé forest dynamics". PI: Ricardo de Lima, Centre for Ecology, Evolution and Environmental Changes, Universidade de Lisboa. The Rufford Foundation 12.517€.
- **National funding as principal investigator: 2014-2018** "**SEEDS -** Closing the seed dispersal loop: How does seed dispersal affect plant population structure at the global, regional and local scales?" Centre for Functional Ecology, University of Coimbra. Fundação para Ciência e a Tecnologia (IF/00441/2013) 50.000€.
- **National funding as team member: 2018-2020** "**BatPine** - Promoting the resilience of ecosystem services to climate change: a case study with pine plantations, pine processionary moth and bats?" PI: Hugo Rebelo, ICETA – Universidade do Porto. P2020 (31731) 223.869€.
- **National funding as team member: 2016-2019** "**INVADER -** Ferramentas INoVAdoras para Detectar Espécies invasoras e agentes de contRolo biológico". PI: Elizabete Marchante, Centre for Functional Ecology, University of Coimbra. Fundação para Ciência e Tecnologia (PTDC/AAG-REC/4896/2014) 199.922€.
- **National funding as team member: 2013-2015** "**GORONGOSA** - Exploring biodiversity and ecological networks for forest conservation and restoration in Gorongosa National Park – Mozambique" PI: Susana Rodriguez-Echeverria, Centre for Functional Ecology, University of Coimbra. Fundação para Ciência e Tecnologia (PTDC/BIA-BIC/4019/2012) 180.972€.
- **National funding as team member: 2013-2015** "**INVADER-B** - Invasive plant species management in Portugal: from early detection to remote sensing and biocontrol of *Acacia longifolia*" PI: Elizabete Marchante, Centre for Functional Ecology, University of Coimbra. Fundação para Ciência e Tecnologia (PTDC/AAG-REC/4607/2012) 199.983€.

Post-graduate supervision

- Post-Docs: Patricia Mateo-Tomás; Beatriz Rumeu Ruiz; Sérgio Antunes Timóteo
- PhD students: Alba Costa Lorenzo; Helena Raposeira; Sasha Vasconcelos; Francisco Lopez-Nuñez; José Miguel Costa; Marta Correia; Luis Pascoal da Silva;
- MSc students: Filipa Mendes; Ana Coelhio; Mauro Nereu; Mariana Valente; Sérgio Ribeiro; Sanjay Sonet; Jioana Cruz.

Outreach

Television

2018 news piece on the TV program "Olhar o Mundo" broadcasted by RTP3 and RTP Africa – 5 February.

2017 Invited scientist on the TV program "Sociedade Civil" broadcasted by RTP2 and RTP Internacional – 6 June.

2015 Invited scientist on the TV program "Mentes que Brilham" broadcasted by Porto Canal – 14 April.

2014 Communication of the Gorongosa project on the TV show "Praça da Alegria" broadcasted by RTP1 – 2 February.

2002 Participation on the TV program "Olhares Cruzados" broadcasted by RTP 2 – 26 November.

Radio

2017 Participation in the series of science outreach podcasts "Todas as plantas têem nome" broadcasted by the Rádio Universidade de Coimbra and published in the journal Diário de Coimbra.

2015 Invited scientist on the radio program "Dias do Futuro" broadcasted by the national station Antena1 – 13 March.

Scientific & Academic management duties

Member of the scientific committee ("Comissão Ciêntifica") of the Department of Life Sciences, Faculty of Sciences and Technology - University of Coimbra (2017 – present).

Representative of doctorate members in the direction board of the R&D unit Centre for Functional Ecology – University of Coimbra (2016 – present).

Editorial boards

Associate editor of the journal "Functional Ecology", British Ecological Society (2017 – present).

Subject editor of the journal "Web Ecology", European Ecological Federation (2014 – present)

Academic juries

PhD jury – Rocio Castro-Urgal, Sergio Chozas Vinuesa; Cristina Tur Espinosa; Danny Rojas Martí; Beatriz Rumeu Ruiz;

Steering committees & Project evaluation boards

External evaluator for the Spanish call for "Unique scientific and technical infrastuctures" 2018

Evaluator for the Marie Skłodowska-Curie Individual Fellowships Europe (H2020-MSCA) 2016, 2017

External project evaluator for the Netherlands Organisation for Scientific Research (NOW) 2016

External project evaluator for the French Agence Nationale de la Recherché (ANR) 2014, 2015

External project evaluator for the Spanish Ministerío de Economia y Competitividad, 2013

IUCN Red List Authority for Birds and Species Survival Commission – Expert member (2012-present).

Organization of scientific meetings

2019 "X Portuguese Ornithological Congress - SPEA". 1-5 March, Peniche, Portugal.

2019 "First Iberian Ecological Society Meeting CSIE2019". 4-7 February, Barcelona, Spain.

2016 "Network approaches to island biology", 2nd Island Biology meeting. 18-22 July, Terceira – Azores, Portugal.

2015 4th Iberian Ecological Meeting: "Ecology and the Societal challenges". 16-19 June, Coimbra – Portugal.

2014 British Ornithologists Union meeting: "Birds in the entangled bank: advances in food-web theory and practice". 19 November Peterborough – UK.

2013 1st International Symposium on "Ecological Networks: Delving into the architecture of biodiversity". 23-25 October, Coimbra – Portugal.

Teaching (157 hours on 8 courses):

- **From 2016 to 2017** Advanced class on "Ecological networks" integrated in the "Lab and field course II" of the European Master in Applied Ecology, Department of Life Sciences, University of Coimbra – Lousã (2x8h). Coord.: José Paulo Sousa
- **From 2015 to 2017** Invited lecturer in the advanced course on "Tropical Ecology" of the master in Conservation Biology, Faculty of Sciences, University of Lisbon (3x1h). Coord.: Jorge Palmeirim.
- **From 2015 to 2017** Invited lecturer in classes of Bioinformatics – Degrees in Biology and Biochemistry, Department of Life Sciences, University of Coimbra (3x2h). Coord.: João Loureiro.
- **From 2012 to 2018** Invited lecturer in the advanced course on "Conservation and Management of Biodiversity" of the European Master in Applied Ecology, Department of Life Sciences, University of Coimbra (7x12h). Coord.: Jaime Ramos.
- **2004** Assistant lecturer in classes of "Entomology", Degree in Biology, School of Biological Sciences, University of Bristol, UK (1x20). Coord.: Richard Wall.
- **2005** Assistant lecturer in classes of "Science and success", Degree in Biology, School of Biological Sciences, University of Bristol, UK (1x10h). Coord.: Jane Memmott.

Weini Huang

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Queen Mary University of London
Mile End Road, London, E14NS
Email weini.huang@qmul.ac.uk

Scientific career and Education

01/2018- **Lecturer in Mathematical Biology**, School of Mathematical Sciences, Queen Mary University of London.

Principal Investigator, Theoretical Biology Lab, College of Life Science, Sun Yat-Sen University, Guangzhou, China.
Research topic: tumour and resistance evolution, predator-prey & host-parasite coevolution, stability of ecological communities.

03/2016-12/2017 **Postdoc**, Evolution and Cancer Laboratory, Centre for Tumour Biology, Barts Cancer Institute, Queen Mary University of London.
Research topic: theoretical modelling of tumour dynamics, evolution of drug resistance in ovarian cancer.

11/2012-01/2016 **Postdoc**, Department of Evolutionary Theory, Max Planck Institute for Evolutionary Biology, Plön, Germany.
Research topic: stochastic models on growing populations, impact of trade-offs on population dynamics under coevolution.

10/2008-08/2012 **Dr. rer. nat.**, Research Group for Evolutionary Theory, Max Planck Institute for Evolutionary Biology, Plön, Germany.
Thesis: Evolutionary game dynamics driven by mutations under frequency dependent selection.

09/2005-07/2008 **M.S.**, Ecology, Beijing Normal University, Beijing.

09/2002-03/2005 **B.A.**, English, Beijing Normal University, Beijing (Minor).

09/2001-07/2005 **B.S.**, Information System and Technology, Beijing Normal University, Beijing (Major).

Publications

- **W. Huang***, A. Traulsen, B. Werner, T. Hiltunen, L. Becks, 2017. Dynamical trade-offs arise from antagonistic coevolution and decrease intraspecific diversity. *Nature Communications*, 8:2059.
- A. Baker, **W. Huang**, et al., 2017. Robust RNA-based in situ mutation detection delineates colorectal cancer subclonal evolution. *Nature Communications*, 8:1998.
- A. Amado, L. Fernández, **W. Huang**, F. F. Ferreira, P. R. A. Campos*, 2016. Competing metabolic strategies in a multilevel selection model. *Royal Society Open Science*, 3:160544.
- **W. Huang***, P. R. A. Campos*, V. M. de Oliveira, F. F. Ferreira*, 2016. A resource-based game theoretical approach for the paradox of the plankton. *PeerJ*, 4:e2329.
- **W. Huang**, C. Hauert, A. Traulsen*, 2015. Stochastic game dynamics under demographic fluctuations. *Proceedings of the National Academy of Sciences*, 112:9064-9069.
- A. Amado, **W. Huang**, P. R. A. Campos, F. F. Ferreira*, 2015. Learning process in public goods games. *Physica A: Statistical Mechanics and its Applications.*, 430:21-31.
- **W. Huang**, B. Werner, A. Traulsen*, 2012. The impact of random frequency-dependent mutations on the average fitness. *BMC Evolutionary Biology*, 12:160.
- **W. Huang**, B. Haubold, C. Hauert, A. Traulsen*, 2012. Emergence of stable polymorphisms driven by evolutionary games between mutants. *Nature Communications*, 3:919.
- X. Luan*, G. Sun, Y. Qu, **W. Huang**, D. Li, S. Liu, B. Wu, 2012. Prioritizing biodiversity in conservation planning based on C-Plan: a case study from northeast China. Acta *Acta Ecologica Sinica*, 32(3):715∼722 (In Chinese with English abstract available).
- **W. Huang**, A. Traulsen*, 2010. Fixation probabilities of random mutants under frequency dependent selection. *Journal of Theoretical Biology*, 263(2):262∼268.
- X. Luan*, **W. Huang**, D. Li, M. Liu, 2009. Identification of hotspots and gaps for biodiversity conservation in the Northeast of China based on a systematic conservation planning methodology. *Acta Ecologica Sinica*, 29(1):144∼150 (In Chinese with English abstract available).
- L. Zhang*, F. Liu, **W. Huang**, J. Song, 2008. Occurrence of Nematode parasites in raptors in Beijing, China. *Journal of Raptor Research*, 42(3):204∼209.

* corresponding authors

Reviewer Activities

Journal of Theoretical Biology, Physical Review Letters, Physical Review E, PNAS, Proc. Royal Soc. B, Proc. Royal Soc. Interface, PLoS Computational Biology, Nature Communications, American Naturalist, BMC Evolutionary Biology, Annals Of Oncology, Gene, Evolution Letters, Evolutionary Applications

FWO, Research Foundation - Flanders in Belgium

Awards and scholarships

2016 The 6th Integrated Mathematical Oncology Workshop "Resistance", Research team award, Moffit Cancer Center, Tampa, US

2016 1000 talented youth program, Ministry of Education, China

2014 Best presentation award, ECMTB 2014, European Conference on Mathematical and Theoretical Biology, Gothenburg, Sweden.

2012-2015 Postdoc fellowship, Max Planck Institute, Germany

2008-2012 PhD scholarship, China Scholarship Council

2008-2009 International fellowship, American Association of University Women Educational Foundation

Organized workshops

07/2017 Foresting cancer evolution symposium, Mathematical Models in Ecology and Evolution, London, Germany.

10/2010 First workshop Gen-AG 'evolutionary genetics': from genes to ecosystems, German Society of Genetics, Kappelsberg, Germany.

Presentations

Invited

- 05/2018, 15th Annual Internal Symposium at the Max Planck Institute for Evolution Biology - Aquavit 2018, Plön, Germany (Guest scientist in the best presentation award committee; Talk: Evolutionary dynamics with tradeoffs).
- 04/2018, Xiamen University, School of Mathematics, Xiamen, China (Talk: Spatial patterns in cancer evolution).
- 12/2017, East China Normal University, School of Environmental Science, Shanghai, China (Talk: From species coevolution to cancer evolution).
- 10/2017, DynaTrait International Conference, Hannover, Germany (Talk: Dynamical trade-offs under coevolution).
- 03/2015, Graduate school of Gulbenkian Institute of Science, Lisbon, Portugal (Mini-course: Evolutionary Game Theory as one part of the Evolution course,1 lecture, 3h/lecture)
- 10/2014, Center for Models of Life, Niels Bohr Institute, Copenhagen, Denmark (Talk: Stochastic game dynamics in populations with changing size).
- 02/2014, Summer school of Department of Physics, Federal University of Pernambuco, Recife, Brazil (Mini-course: Evolutionary Game Theory, 3 lectures, 1.5h/lecture)
- 05/2011, International symposium for biodiversity and theoretical ecology, Guangzhou, China (invited talk: Polymorphisms emerges from evolutionary mutant games).

Others

- 12/2017, The International Society for Evolution, Ecology and Cancer, Tempe, US (Talk: Exploring the evolution of cancer and therapy resistance by theory and experiments).
- 07/2017, Mathematical Models in Ecology and Evolution, London, UK (Talk: Revealing mutant fitness and timing by spatial mixing of sub-clones in tumour).
- 07/2016, Evolutionary Biology and Ecology of Cancer Summer School, Cambridge, UK (poster).
- 05/2015, Modelling Biological Evolution: Linking Mathematical Theories with Empirical Realities, Leicester, United Kingdom (Talk: Modelling frequency dependent interactions between types under demographic fluctuations).
- 06/2014, European conference on mathematical and theoretical biology, Gothenburg, Sweden (Two talks: 1. Evolutionary game dynamics under demographic fluctuations; 2. A resource-based model of multi-species competitions under mutations).
- 08/2013, Models in Population Dynamics and Ecology, Osnabrück, Germany (Talk: Stable polymorphisms driven by evolutionary games between mutants).
- 03/2013, 77th annual meeting of the DPG(German physics society) and DPG spring meeting, Regensburg, Germany (Talk: Evolutionary game dynamics under random mutations).
- 07/2011, Summer school on dynamical models in life sciences organized by European society for mathematical and theoretical biology, Evora, Portugal (poster).
- 09/2009, Evolution of cooperation: models and theories, Laxenburg, Austria (poster).

Field work experiences

08/2006-10/2006 Population survey of elephants in the tropical forests of Xishuanbanna, Yunnan Province, China

07/2006-08/2006 Bird survey in the north-west mountain area in Sichuan Province, China

Other experiences

10/2007 Conservation International, Investigation of illegal trade of Chinese Pangolin (*Manis pentadactyla*) in Jiangxi provionce, China

04/2005-06/2006 Roots & Shoots (The Jane Goodall China), Volunteer & group leader of animal welfare and room enrichment team, Beijing Zoo

09/2005-02/2006	Beijing Raptor Rescue Center, Volunteer & keeper
2004	WWF/ Friend of Nature, Volunteer & environment investigator

Curriculum Vitae

Personal information

Name: Melian, Carlos

Researcher ID: C-2146-2008 ORCID: 0000-0003-3974-6515 SCOPUS: 55220360000

Date of birth: 23/08/1969

Nationality: Spanish

URL for web site: http://www.eawag.ch/en/aboutus/portrait/organisation/staff/profile/carlos-melian/show/

Education

2005 PhD in Biology and Environmental Sciences.
Department of Ecology, University of Alcalá, Madrid, and Biological Doñana Station, Seville, Spain. Supervisor: Prof. Jordi Bascompte.

Professional Experience

2015-Present Lecturer, University of Bern, Switzerland.
2010-Present Group leader (Tenured since end 2015),
Swiss Federal Institute of Aquatic Science and Technology, Switzerland.

Fellowships

2008-2010 **Postdoctoral Fellow**, Microsoft Research Ltd. UK, (Unrestricted Gift).

2005-2007 **Postdoctoral Fellow**, National Center of Ecological Analysis and Synthesis, University of California Santa Barbara, USA.

2001-2004 **PhD. Fellowship**, Spanish Ministry of Science and Technology, Spain

1999 **Student research fellowship**, Ministry of Foreign Affairs. Spanish Agency for International Cooperation, Spain and University of Granma, Bayamo, Cuba.
Title: Linking Agroecology, Conservation and Theoretical Ecology.

1997 **Student research fellowship,** Department of Ecology, Alcala University, Spain
Title: Complex Systems: Theory and its Application to Ecology.

Organization of international scientific meetings

2016 International exploratory workshop “Interactions on evolutionary trees”,
Swiss National Science Foundation, Switzerland.

2008 Workshop “Theoretical ecology: Big questions and challenges”. National Center for Ecological Analysis and Synthesis (NCEAS). Univ. California Santa Barbara, USA.

Supervision and co-supervision of Masters, PhD students, and Postdocs

PhD students

2012-2015 (PhD) student Simone Fontana with Dr. Francesco Pomati
Swiss National Science Foundation Grant 31003A-144162. Switzerland.

2**010-2014** (PhD) PhD. Student Francisco Baldo with Dr. Pilar Drake
PhD. Fellowship, Spanish Ministry of Science and Technology, Spain.

Postdocs

2017 Gian Marco Palamara (Lab discretional funding)

2015 Ayana Martins (BEPE fellowship, Brazil)

2013-2015 Charles N. de Santana (SNSF 31003A_144053)

2013-2014 Jan Klecka (SCIEX 12.327)

2013-1014 Gregor Kalinkat (German Academic Exchange Service, DAAD)

2012 Josephine Rodriguez (Lab discretional funding)

Editorial and Boards

2011-Present Subject editor ***Oikos***

2018 Recommender for PCI Ecology

Teaching activities

2013-present “Introduction to literature analysis”
Institute of Ecology and Evolution, Uni. Bern, Switzerland.

2013 Mini-course “Speciation models” to undergraduates and PhD students. Eawag, Kastanienbaum, Switzerland.

2012 “Evolutionary biodiversity dynamics” with Prof. Ole Seehausen. Institute of Ecology and Evolution, U. Bern, Switzerland.

2007-2009 “Kids Do Ecology” program. National Center for Ecological Analysis and Synthesis (NCEAS). Univ. California Santa Barbara, USA.

Invited talks (top 5 highlighted)

2016 ***MEDECOS conference*** Seville, Spain
"Human driven scenarios for evolutionary and ecological changes"
Keynote speaker symposia
Title: Interdependent Networks in Ecology and Evolution

2015 ***Mathematical models in Ecology and Evolution conference***, Paris, France
Title: Eco-evolutionary diversification dynamics.

2013 ***INTECOL meeting***, London, UK
Title: Connecting diversification and biodiversity dynamics across spatial scales, INTECOL meeting, London, August, UK.

2012 **Web of life conference**, University of Montpellier, France.
Title: Eco-evolutionary dynamics of individual-based networks,

2011 ***ESA, Ecological Society of America Meeting***, Austin, Texas, USA.
Title: Testing adaptive and non-adaptive radiations and biodiversity patterns in supernetworks,

Publication summary

I have 29 publications in international peer reviewed journals, h-index: 18, i-10 index: 25, total citations: 3320 (Google scholar May 29, 2018). I have 17 papers as first/senior author with a mean 5YIF equals to 8.924 of which 13 are without my PhD supervisor with a mean 5YIF equals to 7.3611.

Grant summary

2016 **Melián, C. J.**, Matthews, B., Seehausen, O., and Harmon, L. J.
Granted: Swiss National Science Foundation, International exploratory workshops
Title: Interactions on Trees.
Period: 1 Week, 21,000 CHF.

2014-2015 Kalinkat, G., and **Melián, C. J.** (PI)
Granted: German Academic Exchange Service (DAAD). Germany
Title: Analysing the interplay between allometric constraints and intraspecific trait variation to predict food web dynamics.
Period: 6 Months, 17,694€.

2014-2015 **Melián, C. J.** (PI) and Klecka, J.
Granted: Swiss National Science Foundation, International Coop. Switzerland.
Title: Modeling the effects of predator phenotype on food web structure.
Period: 3 Months, 9,510 CHF.

2012-2015 **Melián, C. J.** (PI)
Granted: Swiss National Science Foundation, Division III. Switzerland.
Title: A theory for next-generation food web data.
Period: 1 and a half Years (Postdoc), 161,000 CHF.

2012-2014 **Melián, C. J.** (PI) , Klecka, J., Křivan, V., and Altermatt, F.
Granted: Sciex, Switzerland.
Title: Food webs in space: integrating metacommunity and food web research.
Period: 1 Year, 92,900 CHF

Prices and awards

2008-2010 **Postdoctoral Fellow** granted by Microsoft Research Ltd. (Unrestricted Gift).
Place: NCEAS, University of California, Santa Barbara, USA.
Title: Unifying Theories of Molecular, Community and Network Evolution.
Period: 2 Years, 200,000$.
Project page: https://www.nceas.ucsb.edu/projects/12306

Peer-reviewed articles (five last years 2014-2018)

1. **Melián, C. J.**, Matthews, B., de Andreazzi, C. S., Rodriguez, J. P., Harmon, L., and Fortuna, M. A. (2018). Deciphering the interdependence between ecological and evolutionary networks. *Trends in Ecology and Evolution* doi.org/10.1016/j.tree.2018.04.009. [IF:15.268]

2. de Andreazzi, C. S., *Guimaraes, P., and ***Melián, C. J.** (2018). Eco-evolutionary feedbacks promote fluctuating selection and long-term stability of antagonistic networks. *Proceedings of the Royal Society B*, 285:20172596. *Last co-authors. [IF:4.94]

3. Pellissier, L., Albouy, C., Bascompte, J., Farwig, N., Graham, C., Loreau, M., Maglianesi, M. A., **Melián, C. J.**, Pitteloud, C., Roslin, T., Rohr, R., Saavedra, S., Thuiller, W., Woodward, G., Zimmermann, N. E., and Gravel, D. (2017). Comparing species interaction networks along environmental gradients, *Biological Reviews*, 93:785-800. [IF:11.615]

4. De Laender, F., Rohr, J. R., Ashauer, R., Baird, D. J., Berger, U., Eisenhauer, N., Grimm, V., Hommen, U., Maltby, L., **Melián, C. J.**, Pomati, F., Roessink, I., Radchuk, V., and Van den Brink, P. J. (2016). Reintroducing environmental change drivers in biodiversity–ecosystem functioning research. *Trends in Ecology and Evolution*, 31:905-915. [IF:15.268]

5. Leprieur, F., Descombes, P., Gaboriau, T., Cowman, P. F., Parravicini, V., Kulbicki, M., **Melián, C. J.**, de Santana, C. N. Heine, C., Mouillot, D., Bellwood, D. R., and Pellissier, L. (2016). Plate tectonics drive tropical reef biodiversity dynamics. *Nature Comm.*, 7:11461. [IF:12.124]

6. *Fronhofer, E., *Klecka, J., **Melián, C. J.**, and Altermatt, F. (2015). Condition-dependent movement and dispersal in experimental metacommunities. *Ecology Letters*, 18:954-963. *First co-authors. [IF:9.449]

7. Hines, J., van der Putten, W. H., De Deyn, C. B., Wagg, C., Voigt, W., Mulder, C., Weisser, W. W., Engel, J., **Melián, C. J.**, Scheu, S., Birkhofer, K., Ebeling, A., Scherber, C., and Eisenhauer, N. (2015). Towards an integration of biodiversity–ecosystem functioning and food web theory to evaluate relationships between multiple ecosystem services. *Advances in Ecological*

Research 53:161-199. [IF:5.056]

8. **Melián, C. J.**, Seehausen, O., Fortuna, M. A., Eguíluz, V., and Deiner, K. (2015). Diversification and biodiversity dynamics of hot and cold spots. *Ecography*, 38:393-401. [IF:4.902]

9. Sanz-Aguilar, A., Jovani, R., **Melián, C. J.**, Pradel, R., and Tella, J. L. (2015). Multievent capture-recapture analysis reveals individual foraging specialisation in a generalist species. *Ecology*, 96:1650–1660. [IF:4.809]

10. ***Melián, C. J.**, *Křivan, V., Altermatt, F., Stary, P., Pellissier, L., and De Laender, F. (2015). Dispersal dynamics in food webs. *The American Naturalist*, 185:157-168. *First co-authors. [IF: 4.167]

11. **Melián, C. J.**, Baldó, F., Matthews, B., Vilas, C., González-Ortegón, E., Drake, P., and Williams, R. J. (2014). Intraspecific trait variation and diversity in food webs. *Advances in Ecological Research*, 50:207-241. [IF:5.056]

12. Moya-Laraño, J., Bilbao-Castro, J. R., Barrionuevo, G., Ruiz-Lupión, D., Casado, L. G., Montserrat, M., **Melián, C. J.**, and Magalhães, S. (2014). Eco-evolutionary spatial dynamics: rapid evolution and isolation explain food web persistence. *Advances in Ecological Research*, 50:75-143. [IF:5.056]

13. *De Laender, F., ***Melián, C. J.**, Bindler, R., Van den Brink, P., Daam, M., Roussel, H., Juselius, J., Verschuren, D., and Janssen, C. (2014). The Contribution of intra- and interspecific tolerance variability to biodiversity changes along toxicity gradients. *Ecology Letters*, 17:72–81. *First co-authors. [IF:9.449]

Scientific Exchanges
Self-declaration

Self-declaration

This self-declaration certifies that:

- the Scientific Exchange submitted by the applicant is not directly linked to any other activity financed in the same timeframe by the SNSF under another funding instrument;
- the visit that the applicant has applied for under the Scientific Exchanges scheme will not take place during a regular sabbatical to which the applicant is entitled every few years as an employee. (Note: this only applies for visits and for Swiss researchers going abroad.)

I, Carlos Melian, herewith confirm that the statements above are correct.

Name of the applicant	Carlos Melian
Date of birth	23/08/1969
Employer of the applicant	Eawag
(Home institution, city and country)	Eawag, Dubendorf, Switzerland

Comments

Date	May 31, 2018
Name of the applicant	Carlos Melian

Signature of the applicant ....

		Resp.	Room		Resp.	Room		Resp.	Room	Date	Resp.	Room	Date	Resp.	Room
	Monday 15.10.2018			Tuesday 16.10.2018			Wednesday 17.10.2018			Thursday 18.10.2018			Friday 19.10.2018		
0700 - 08.00	Breakfast	Nadja	BH Kitchen	Breakfast	Nadja	BH Kitchen	Breakfast	Nadja	BH Kitchen	Breakfast	Nadja	BH Kitchen	Breakfast	Carlos	BH Kitchen
900	Welcome	Carlos	BH 1st Floor	Data-Models	all		Intro	Francois		Intro	Julia		Intro	Catherine	
1000 **1100** **1200**	1st session talks: Empirical patterns	all	BH 1st Floor	2nd session talks: Empirical data	all	SH / BH / BH1st floor	3rd session: Theoretical approaches	Group	SH in the back / BH / BH 1st floor	Writting session I	all	SH in the back / BH / BH 1st floor	Writting session II	all	SH in the back / BH / BH 1st floor
12.00 - 13.00	Lunch	Carlos	BH Kitchen	Lunch	Carlos	BH Kitchen							Lunch	Carlos	BH Kitchen
1300	Outlook	Lutz	Seeheim	Outlook	Cecilia	SH	Lunch	Carlos	BH Kitchen	Lunch	Carlos	BH Kitchen	Writting session III	all	SH in the back / BH / BH 1st floor
1400 **1500**	Discussion	Cecilia	Seeheim	3rd session: Merging empirical patterns and models	all	SH / BH Kitchen	Seminar	Catherine Graham	SH	Excursion	Carlos				
1600	Breakout groups	Carlos	SH / BH Kitchen												
Buil				Hiking	all	SH / BH Kitchen	Database building	Group	SH in the back / BH / BH 1st floor				Final discussion / planning	all	SH in the back / BH / BH 1st floor
1800	Lucerne	Carlos		No activity	all		BBQ	Carlos (Grill)	SH	Dinner	Carlos		Dinner Strandbad Horw	Nadja	
1900	Apero	Carlos													
2000 - 21.00	Dinner Rest. Pfistern, Luzern	Nadja		Dinner Lucerne	all										

Activities at Kastanienbaum

	Monday	Tuesday	Wednesday	Thursday	Friday
Room Booking	Surf Talk Seeheim 10.00 – 12.00	Meeting River Rest. BH 1st floor 15.00 - 16.30	Student Discussion /JC SH in the front 10.15 - 12.30		No activity
	Meeting BH 1st floor 15.30 - 18.00				
Admin presence	Check	Nadja, Patricia	Nadja	Nadja	Michelle

Research part for the proposal of a Scientific Event

Title: Biodiversity Dynamics in Coevolutionary Meta-ecosystems (COMETA)

Carlos J. Melián

1. Scientific aspects

1.1 Introduction of the research subject

Coevolution, the trait change driven by reciprocal selection within and among interacting species, has been invoked as a primary mechanism generating much of Earth's biodiversity (Althoff et al., 2014, Hembry et al., 2014). Two main theories have been proposed to explain the dynamics of coevolutionary diversification. The Fisherian sexual selection theory, which states that female and male traits can lead to sexual isolation and speciation (Fisher, 1930, Pomiankowski and Iwasa, 1998), and the Geographic mosaic theory of coevolution (Thompson, 2005), which suggests that reciprocal changes in traits determining species interaction produce coevolutionary diversification as interspecific interactions control reproductive isolation and speciation. Despite the appeal of such theories for explaining much of Earth's biodiversity, understanding how coevolution of intra- and inter-specific traits associated with ecological interactions alter ecosystem dynamics across heterogeneous landscapes remains elusive. Such knowledge understanding is particularly important in ecosystems with species- and interaction-rich ecological networks because the global change might trigger many species extinctions and the collapse of ecosystems (Thompson, 2014).

In this vein, To address this challenge we will join theoretical and empirical evolutionary ecologists to develop theoretical models that incorporate the empirical patterns of reciprocal trait changes will be critical to improve our predictions on biodiversity dynamics in species- and interaction-rich ecological communities. The aim of the workshop is to integrate hierarchical trait-species-space interaction data with new process-based coevolutionary metaecosystem models. Such integration might facilitate the exploration of patterns of coexistence of co-evolved species, their role in ecosystem function, and the risk of collapse of co-evolved ecosystems across different spatio-temporal scales, which is of critical importance for three reasons. First, the extinction of species caused by human-disturbances is occurring at an unprecedented rate, yet, most models predicting extinction rates often assume that species evolve independently of one another (Thompson, 2014). Second, despite recent advances in eco-evolutionary approaches, studies connecting co-evolving trait distributions within and between species at small scales to investigate the large-scale association between coevolutionary hot spots, biodiversity function and meta-ecosystem dynamics are rare (Doebeli and Ispolatov, 2017). Third, although many experiments and empirical studies have shown convergence of time scales between ecological and coevolutionary processes (Hairston et al., 2005, Fussmann et al., 2007, Schoener, 2011), there is a gap in understanding how ecological and coevolutionary dynamics in species-rich assemblages shape biodiversity and ecosystem function in a spatio-temporal context – i.e. the meta-ecosystem

(Gomulkiewicz et al., 2000, Thompson, 2013).

A meta-ecosystem is a set of local ecosystems connected by spatial flows of energy, materials and organisms across ecosystem boundaries (Loreau et al., 2003). The integration of meta-ecosytem ecology, coevolutionary processes within and between species and ecosystem functioning remains one of the main challenges for theoretical and applied ecologists. This paradigm provides a powerful tool to understand the properties that arise from the spatial coupling of local ecosystems, such as coevolutionary selection mosaics, stabilization of ecosystem processes and indirect interactions at regional scales (Gomulkiewicz et al., 2000, Gravel et al., 2010). We propose a coupled workshop and research visit aiming to integrate empirical datasets, theory and analytical methods on coevolution and meta-ecosystems to bridge the gap between coevolutionary and ecological spatio-temporal dynamics of species- and interaction-rich networks. This integration will help us to advance our understanding of biodiversity dynamics and ecosystem functioning across spatial and temporal scales.

We believe it is the right time for such a workshop. There have recently been many advances in mathematical approaches for studying coevolution (Althoff et al., 2014, Hembry et al., 2014, Doebeli and Ispolatov, 2017). However, most of these approaches are indirectly connected to biodiversity dynamics and meta-ecosystems functioning, and assume that species coevolve in isolated communities. This assumption, which is more associated to mathematical convenience than to biological and natural history knowledge, limits our understanding of the relationships among coevolution, biodiversity dynamics and ecosystem function across spatio-temporal scales in species-rich interacting systems. At the same time, there

During the workshop and research visit we will combine methods and models to incorporate co-evolving trait distributions and species interactions in meta-ecosystems (Gravel et al., 2016, DeLong and Gibert, 2016, Massol et al., 2017, Andreazzi et al., 2018, Melián et al., 2018). Such approaches will consider spatially variable biotic selection and individual migration among heterogeneous patches to generate complex selection mosaics, leading to spatial variation in species traits, and therefore to predictions of the structure of ecological networks in meta-ecosystems. Therefore, we will seek to combine coevolutionary dynamics, meta-ecosystems and interaction networks, by synthesizing spatial and trait-interaction data in species-rich ecosystems to contribute.

1.2 Scientific aims and methods of the workshop

We will focus on implementing analytic tools and testing them with empirical patterns from data sets containing many species, sites and traits (Heleno et al., 2012, Aizen et al., 2016, Frickel et al., 2016, Weinstein and Graham, 2017) (Table 1). We propose bringing together biologists skilled in theoretical ecology, evolutionary biology, and community ecology. We will focus on the following two questions about co-evolution in meta-ecosystems.

Question 1: How does reciprocal trait change influence biodiversity dynamics, species *extinctions and ecosystem function in species-rich meta-ecosystems?*

We aim to integrate the effects of ecological, coevolutionary, and migration dynamics in shaping biodiversity in meta-ecosystems. We will merge the meta-ecosystem perspective (Gravel et al., 2016, Massol et al., 2017, Melián et al., 2018) into existing models of coevolutionary interaction networks (Guimarães et al., 2011, Andreazzi et al., 2017). We will implement scenarios to predict biodiversity distribution across space and time under different functional interactions (*i.e., trait matching and exploitation barriers*) assuming weak and strong reciprocal selection strengths. Metacommunity and meta-ecosystem theory approaches will complement the functional scenarios to explore how spatial processes governed by dispersal (i.e. mass effect) or niche width (i.e. species sorting, patch dynamics) can alter the coevolutionary dynamics within communities by contributing to species trait variation and network structure and stability, and thus ecosystem functioning (Mouquet and Loreau, 2003, Leibold et al., 2004, Logue et al., 2011, Winegardner et al., 2012, Borthagaray et al., 2014 a,b, 2015a, Astegiano et al., 2015a, Laroche et al., 2016, Massol et al., 2017 a, b).

Question 2: How do the spatial arrangement of ecosystems and coevolutionary dynamics interact to facilitate rapid evolutionary changes in meta-ecosystems?

Most interactions between species generate reciprocal selection pressures, and the coevolutionary dynamics that emerges over time varies with the fitness consequences of the ecological interaction (i.e., competitive, mutualistic, or antagonistic) and the architecture of the coevolving interactions between species (Wade 2007, Loeuille and Loreau, 2005, Doebeli 2011, Nuismer et al., 2013, Allhoff and Drossel, 2013, Althoff et al., 2014, Hembry et al., 2014). Yet, how the relationship between species traits and coevolution at local scales and the arrangement of ecosystems at the landscape scale produce rapid biodiversity changes and coevolutionary hot spots is not well understood (Wade, 2007, Doebeli, 2011, Althoff et al., 2014, Hembry et al., 2014). We will run simulations under landscapes with different spatial arrangements and migration scenarios to disentangle the interplay between coevolutionary dynamics at the local and the landscape level.

1.3 Expected results

We will work on an opinion, review or ideas-perspectives paper entitled "*Biodiversity dynamics and ecosystem functioning in coevolutionary meta-ecosystems*" for submission to *Trends in Ecology and Evolution*, *Ecology Letters* or *Biological reviews*. The two questions above will form the core of the discussions at the workshop, and they will help drafting an outline for the paper during the meeting. Prior to the meeting, attendees will provide written perspectives about how to address each of the overarching questions and the trait-species-spatial interaction data they can provide. On the first day, participants will present these ideas based on their own research programs. On subsequent days we will synthesize these perspectives into a single review/opinion and synthesis style paper integrating the datasets to the methods that we will develop and implement during the workshop and following research visit stay.

2. Organizational aspects

2.1 Milestones and program

The main milestones of the workshop will be:

1. Draft the outline for a synthesis paper about Biodiversity dynamics and ecosystem functioning in coevolutionary meta-ecosystems summarizing the main findings obtained during the workshop.

2. Discuss future directions for the field that unify short- and long-term approaches integrating modeling and data.

3. Integrate new modeling frameworks to empirical data and make them available throughout github and related open repositories to facilitate reproducibility of the analysis.

4. Develop new ideas to strengthen future grant proposals and collaborations

2.2 Venue

The workshop will be hosted at Eawag's Center for Ecology, Evolution & Biogeochemistry (CEEB), in Kastanienbaum. This is an ideal venue. All the invited guests can stay on-site in Eawag apartments, directly on the shore of Lake Lucerne. The meeting facilities are very suitable and have been used before for such workshops. We have previously used this location to host several small workshops and student courses with great success. Such events also help to raise our international profile, and, in the past, have led to repeat visits from graduate students and visiting faculty on sabbatical terms.

2.3 Other funding sources

Eawag will provide the necessary infrastructure for the workshop (venue, consumables, IT, etc.). No other funds are needed to support this workshop.

3. Partnership aspects

3.1 Complementarity aspects

We have selected a list of participants that represent a wide range of ages, expertise, and levels of experience. We have identified a mixture of people with (i) skills in mathematical biology, (ii) experience with software development, and (iii) strong conceptual skills in evolutionary biology, meta-ecosystem theory, model systems and field studies.

3.2 Potential for further collaboration

This workshop will lead to new international collaborations. We see great potential in linking the research of the participants to produce new synthesis in merging coevolution, interaction networks and meta-ecosystems. Specifically, deciphering the interplay between coevolution in local ecosystems for antagonistic and mutualistic interactions with the effects of migration under different landscape arrangements is a topic that can move forward the needed synthesis between the ecological and evolutionary processes that shape biodiversity. Furthermore, our understanding of coevolution among many interacting species at different spatial scales and its effects on biodiversity dynamics and ecosystem function is just starting to emerge. Yet, the integration between robust empirical patterns and process-based models, the main aim of this workshop, is currently missing.

Table 1: Empirical data

	Traits (R/C[1])	**Space/Habitats/Distance matrix**	**Species**	**Reference**
Plant-Bird-Reptile	R:No/C:Yes	Yes/Yes/Yes	R:45/C:13	***Heleno et al 2012*** (approx. 5k bird-reptile individual observations)
Plant-Hummingbirds	R:Yes/C:Yes	Yes/Yes(gradient)/Yes	R:40/C:23 Genera	***Weinstein and Graham, 2017*** (approx. 5k hummingbird individual observations)
Plant-Pollinators	R:Yes/C:Yes	Yes/No/Yes	R:96/C:172	***Aizen et al., 2016*** (approx. 6k pollinator individual observations)
Host-virus	R:Yes/C:Yes	No (replicates)/No/No	R:Types/Types	***Frickel et al., 2016***

3.3 References

Aizen, M. A., Gleiser, G., Sabatino, M., Gilarranz, L. J., Bascompte, J., and Verdú, M. (2016). The phylogenetic structure of plant-pollinator networks increases with habitat size and isolation. ***Ecology Letters***, 19:29-36.

Allhoff, K.T. and Drossel, B. (2013). When do evolutionary food web models generate complex networks? ***J. Theor. Biol.***, 334:122–129.

Althoff, D. M., Segraves, K. A., and Johnson, M. T. J. (2014). Testing for coevolutionary diversification: linking pattern with process. ***Trends in Ecology and Evolution***, 29:82-89.

Astegiano, J., Guimarães Jr, P. R., Cheptou, P. O., Vidal, M. M., Mandai, C. Y., Ashworth, L., and Massol, F. (2015). Persistence of plants and pollinators in the face of habitat loss: insights from trait-based metacommunity models. ***Advances in Ecological Research***, 53:201-257.

1 R: resource trait and C: consumer traits

Andreazzi, C. S., Guimarães, P. R., and Melián, C. J. (2018) Eco-evolutionary feedbacks promote fluctuating selection and long-term stability of antagonistic networks, ***Proceedings of the Royal Society B: Biological science***, 285:20172596.

Borthagaray, A. I., Barreneche, J. M., Abades, S., & Arim, M. (2014a) Modularity along organism dispersal gradients challenges a prevailing view of abrupt transitions in animal landscape perception. ***Ecography***, 37:564-571.

Borthagaray, A. I., Arim, M., & Marquet, P. (2014b). Inferring species roles in metacommunity structure from species coexistence networks. ***Proceedings of the Royal Society B***, 281:20141425.

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